FEMB QC Criteria Report

FEMB-IO 1826-1L Quality Control Analysis

Analysis Date: October 2026 **Board Version:** FEMB-IO 1826-1L **Number of Boards Analyzed:** 44
Analysis Method: Statistical analysis using latest passing test per board

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Overview

This report documents the Quality Control (QC) criteria derived from 44 FEMB-IO 1826-1L boards that passed checkout testing at BNL. The criteria are established using statistical analysis with 3σ (99.7% coverage) and 5σ (99.99994% coverage) bounds.

Statistical Methods

- **Mean:** Arithmetic average of all measurements
- **Std:** Standard deviation
- **Median:** Middle value (robust to outliers)
- **MAD:** Median Absolute Deviation (robust measure of spread)
- **3σ Bounds:** Mean ± 3×Std (99.7% of normal distribution) - shown in green
- **5σ Bounds:** Mean ± 5×Std (recommended pass/fail threshold) - shown in red

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1. 128-Channel Parameters

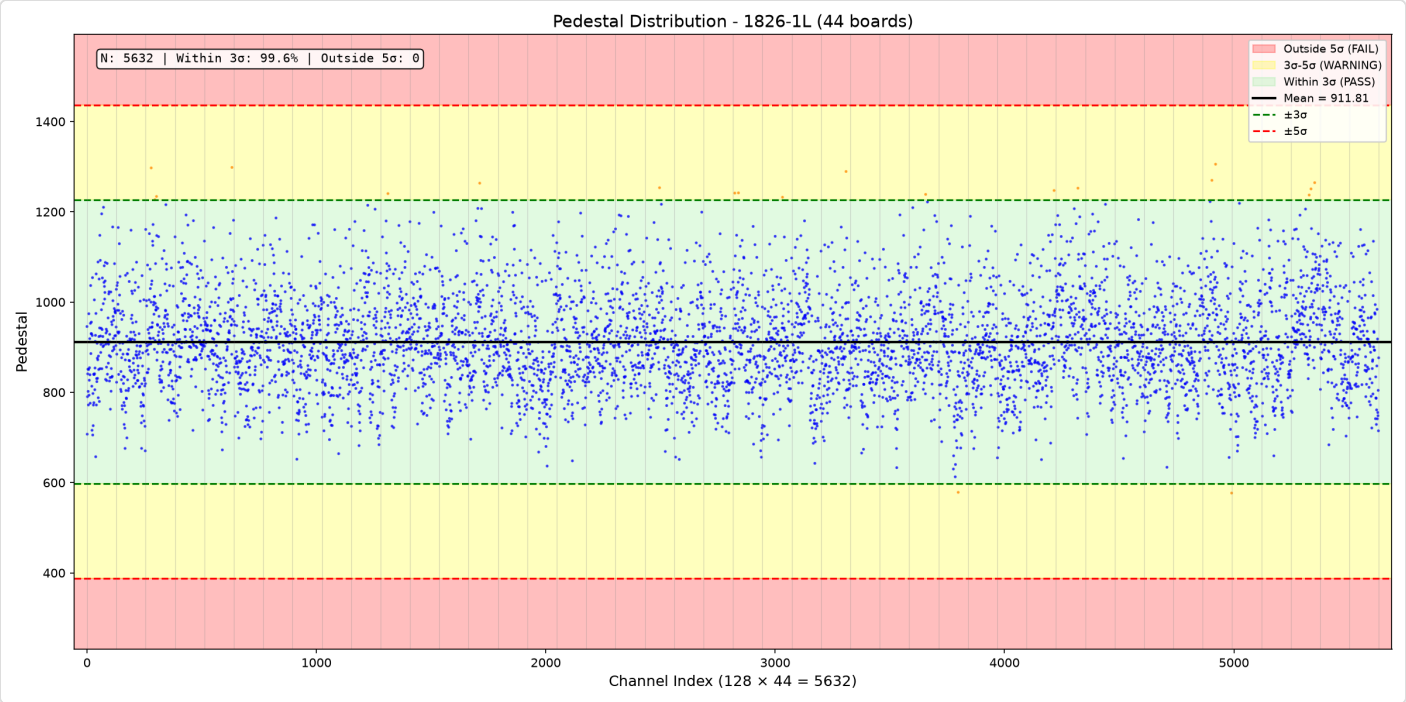
These parameters are measured for all 128 channels on each FEMB board (5632 total data points = 44 boards × 128 channels).

1.1 Pedestal (ped)

Description: The baseline ADC value when no signal is present. Represents the DC offset of each channel's readout electronics.

Statistic	Value
Mean	911.81
Std	104.80
Median	903.32
MAD	67.86
3σ Lower	597.41
3σ Upper	1226.20
5σ Lower	387.81
5σ Upper	1435.80

Interpretation: Pedestal values vary across channels due to manufacturing variations. Values outside 5σ bounds may indicate channel issues.

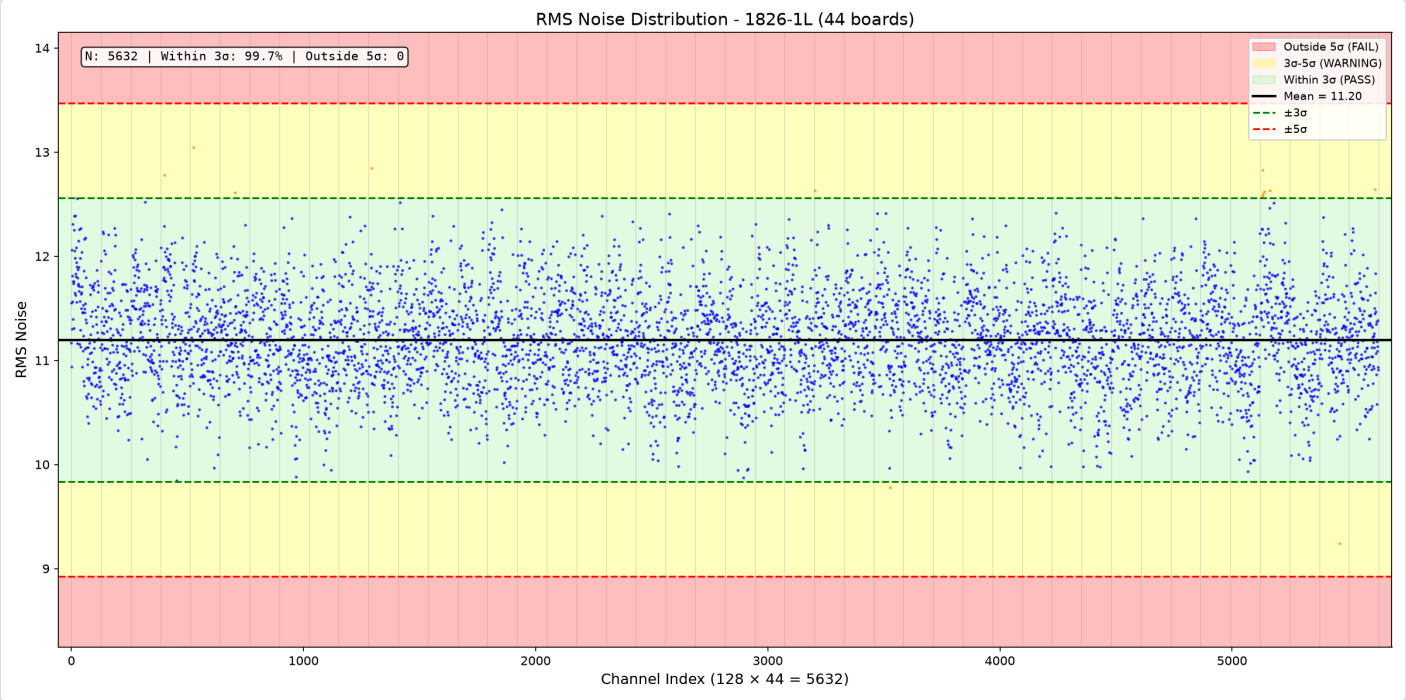


1.2 RMS Noise (rms)

Description: Root Mean Square of the baseline fluctuations. Measures the intrinsic electronic noise of each channel in ADC counts.

Statistic	Value
Mean	11.20
Std	0.45
Median	11.19
MAD	0.30
3σ Lower	9.83
3σ Upper	12.56
5σ Lower	8.93
5σ Upper	13.47

Interpretation: Lower noise is better. High RMS indicates noisy channels that may affect signal quality. Typical good channels have RMS around 11 ADC counts.

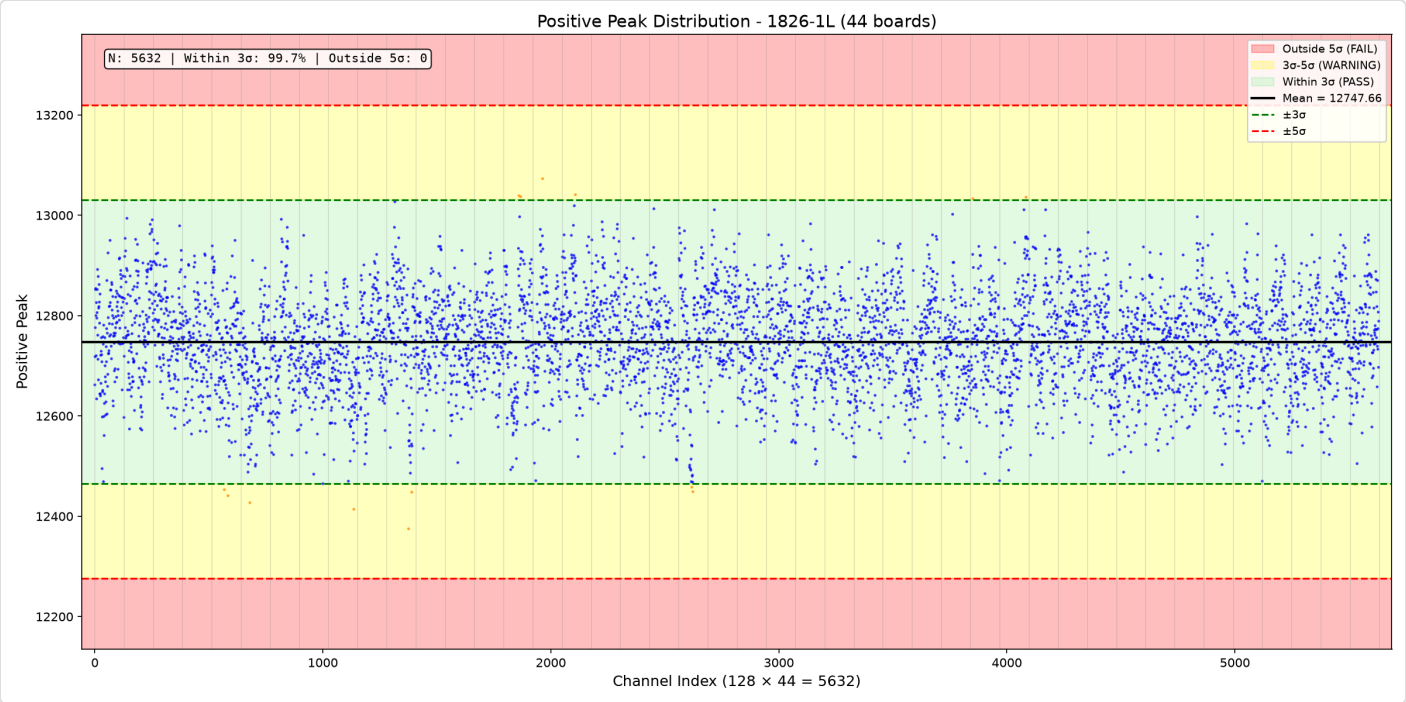


1.3 Positive Peak (PPK_mean)

Description: Mean ADC value of the positive pulse response when injecting a calibration pulse. Measures the channel gain for positive signals.

Statistic	Value
Mean	12747.66
Std	94.28
Median	12749.00
MAD	64.00
3σ Lower	12464.80
3σ Upper	13030.51
5σ Lower	12276.23
5σ Upper	13219.08

Interpretation: Consistent PPK values across channels indicate uniform gain. Large deviations suggest gain calibration issues.

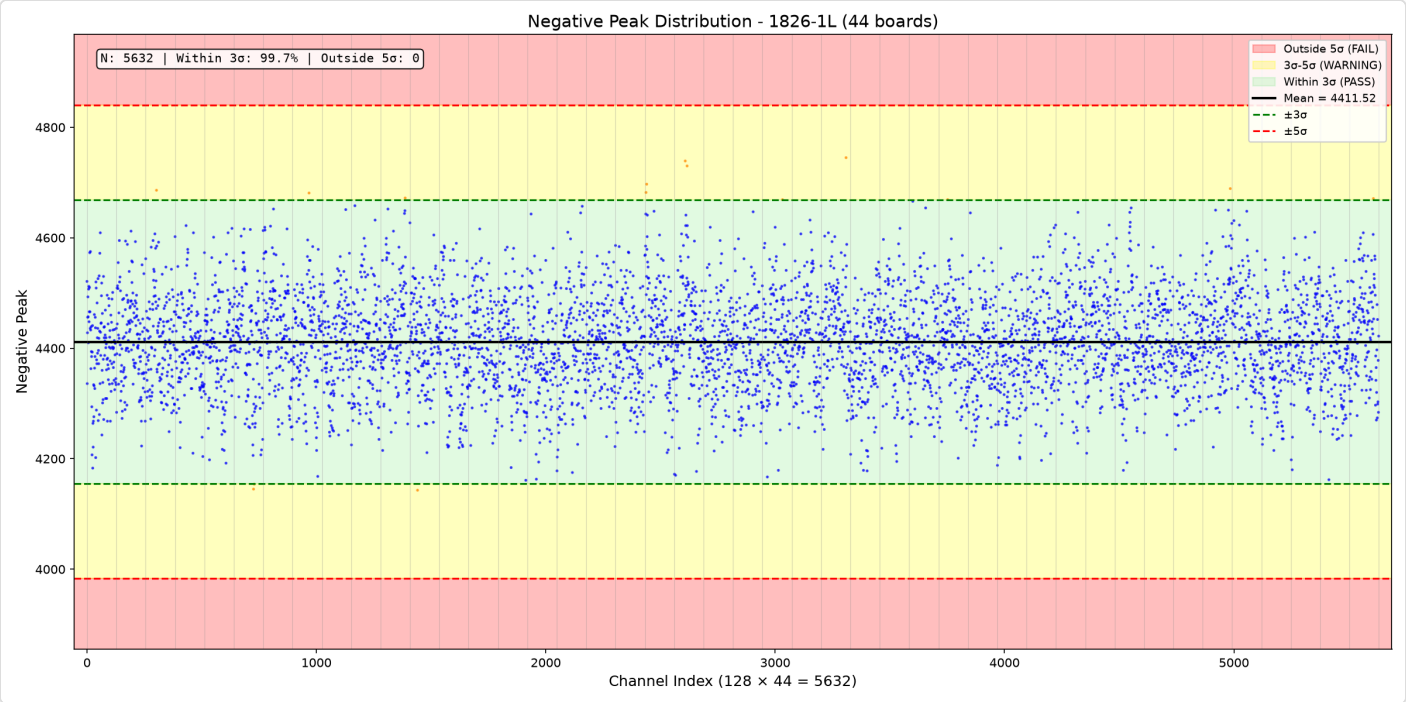


1.4 Negative Peak (NPK_mean)

Description: Mean ADC value of the negative pulse response. Measures the channel response to negative-going signals.

Statistic	Value
Mean	4411.52
Std	85.62
Median	4411.00
MAD	57.00
3σ Lower	4154.66
3σ Upper	4668.39
5σ Lower	3983.41
5σ Upper	4839.64

Interpretation: NPK should be symmetric with PPK relative to baseline. Asymmetry may indicate amplifier nonlinearity.

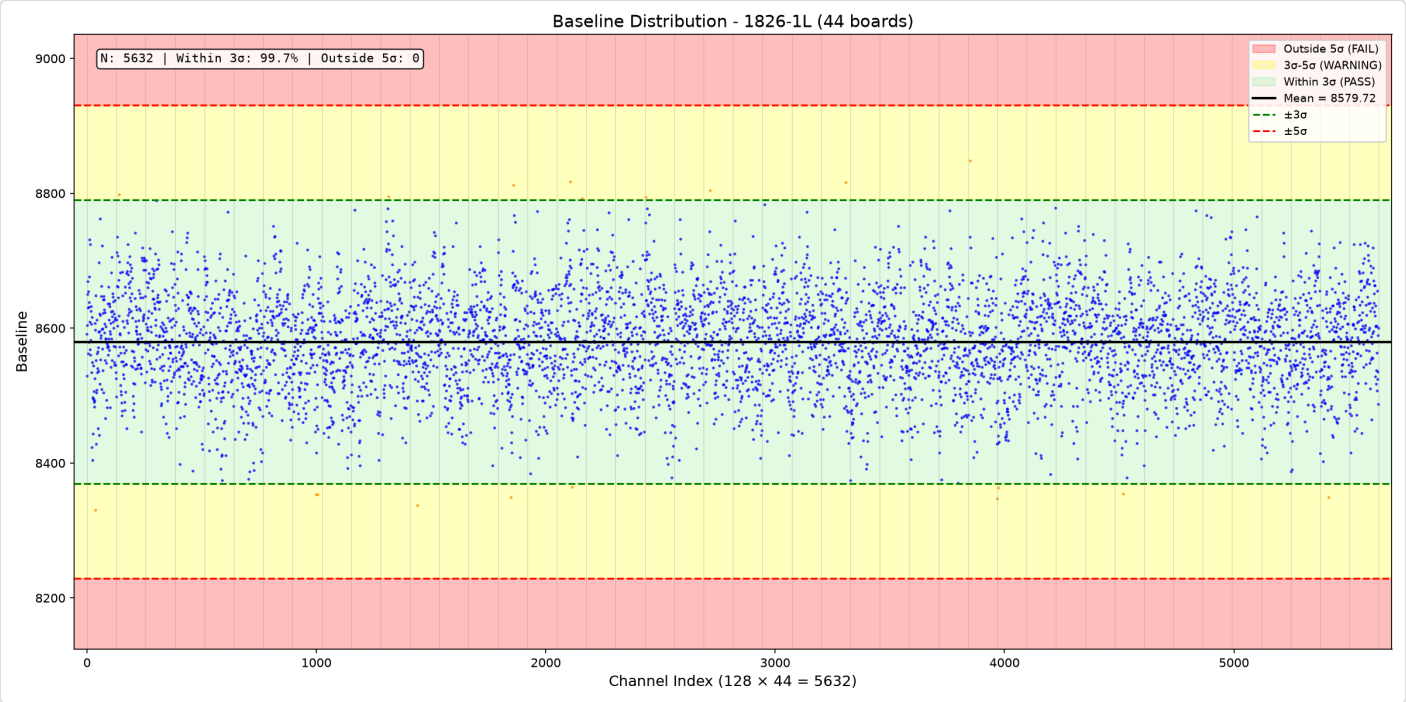


1.5 Baseline (BL_mean)

Description: Mean baseline level during pulse response measurement. Should be centered around mid-range ADC value.

Statistic	Value
Mean	8579.72
Std	70.14
Median	8581.00
MAD	47.00
3σ Lower	8369.29
3σ Upper	8790.16
5σ Lower	8229.00
5σ Upper	8930.45

Interpretation: Baseline provides reference for pulse amplitude measurements. Stable baseline indicates good DC performance.

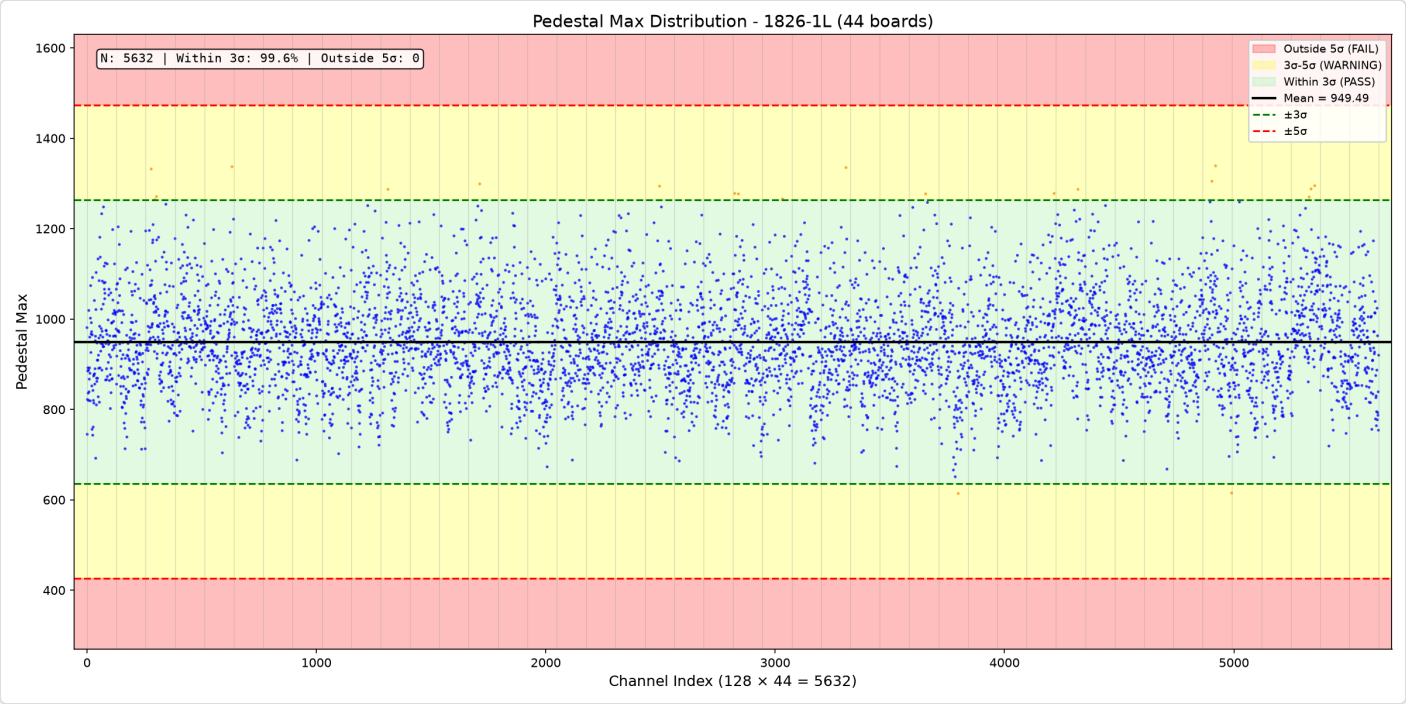


1.6 Pedestal Max (pedmax)

Description: Maximum pedestal value observed during the measurement window. Used to check for pedestal stability.

Statistic	Value
Mean	949.49
Std	104.66
Median	941.00
MAD	68.00
3σ Lower	635.51
3σ Upper	1263.46
5σ Lower	426.19
5σ Upper	1472.78

Interpretation: Large difference between pedmax and pedmin indicates pedestal instability or excessive noise spikes.

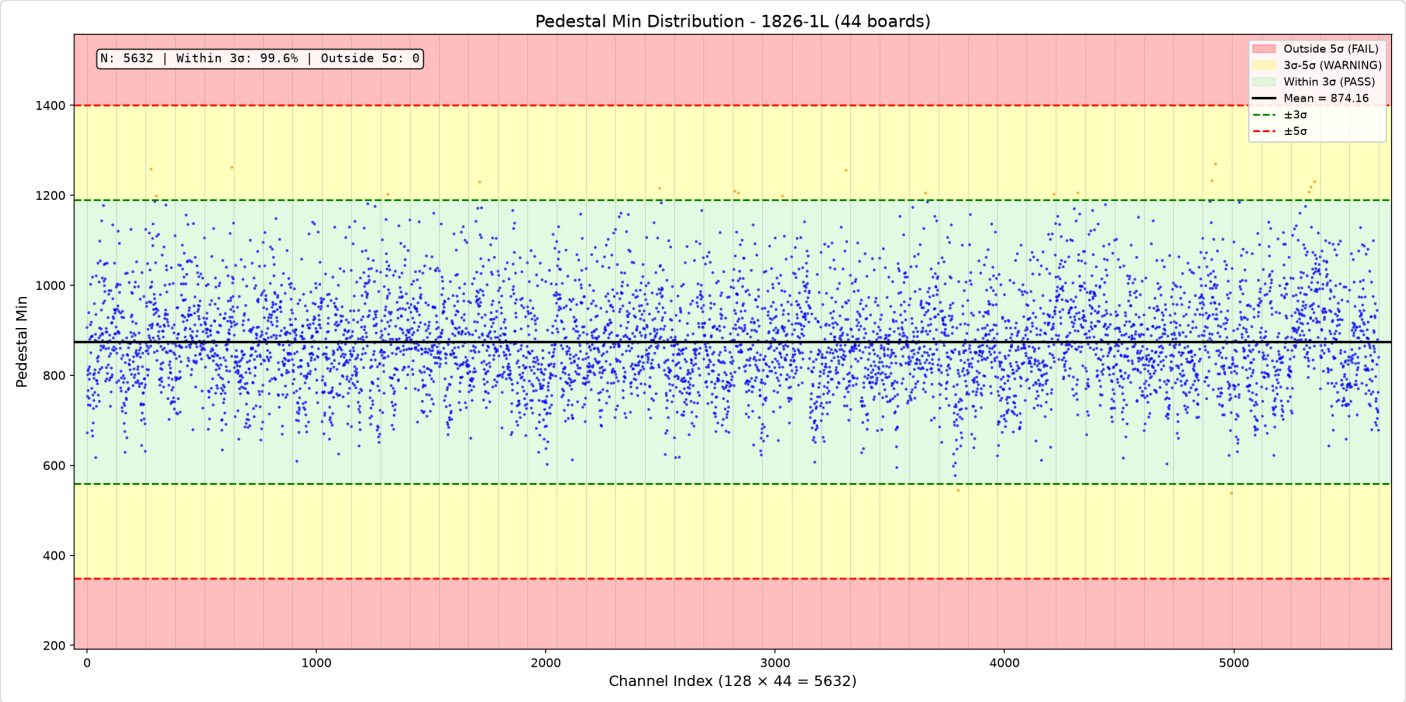


1.7 Pedestal Min (pedmin)

Description: Minimum pedestal value observed during the measurement window.

Statistic	Value
Mean	874.16
Std	105.09
Median	865.00
MAD	68.00
3σ Lower	558.90
3σ Upper	1189.42
5σ Lower	348.73
5σ Upper	1399.60

Interpretation: Together with pedmax, defines the pedestal variation range. (pedmax - pedmin) should be small for stable channels.



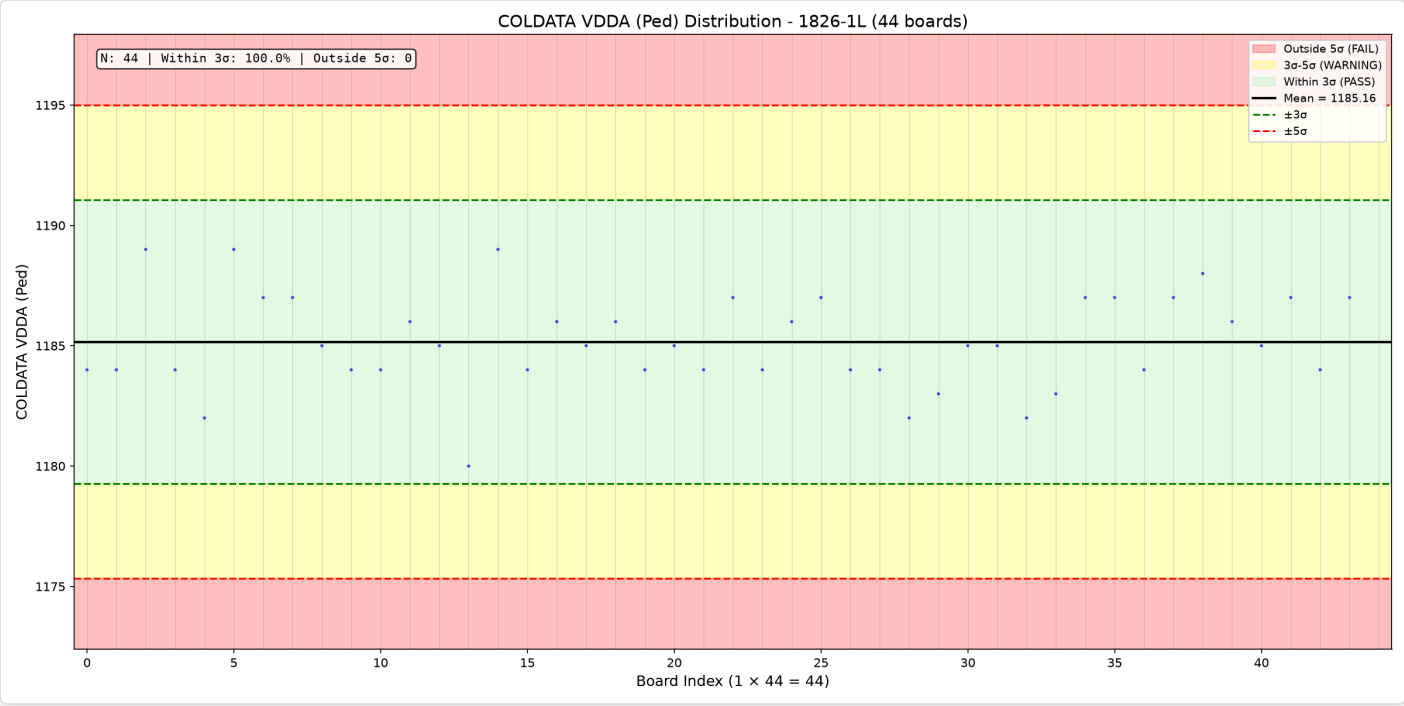
2. LDO Voltage Parameters (Pedestal Configuration)

Low Dropout (LDO) regulator output voltages measured during pedestal check. These monitor the on-board power regulation.

2.1 CDVDDA_ped

Description: COLDATA analog supply voltage (VDDA) during pedestal check. Powers the analog sections of the COLDATA ASIC.

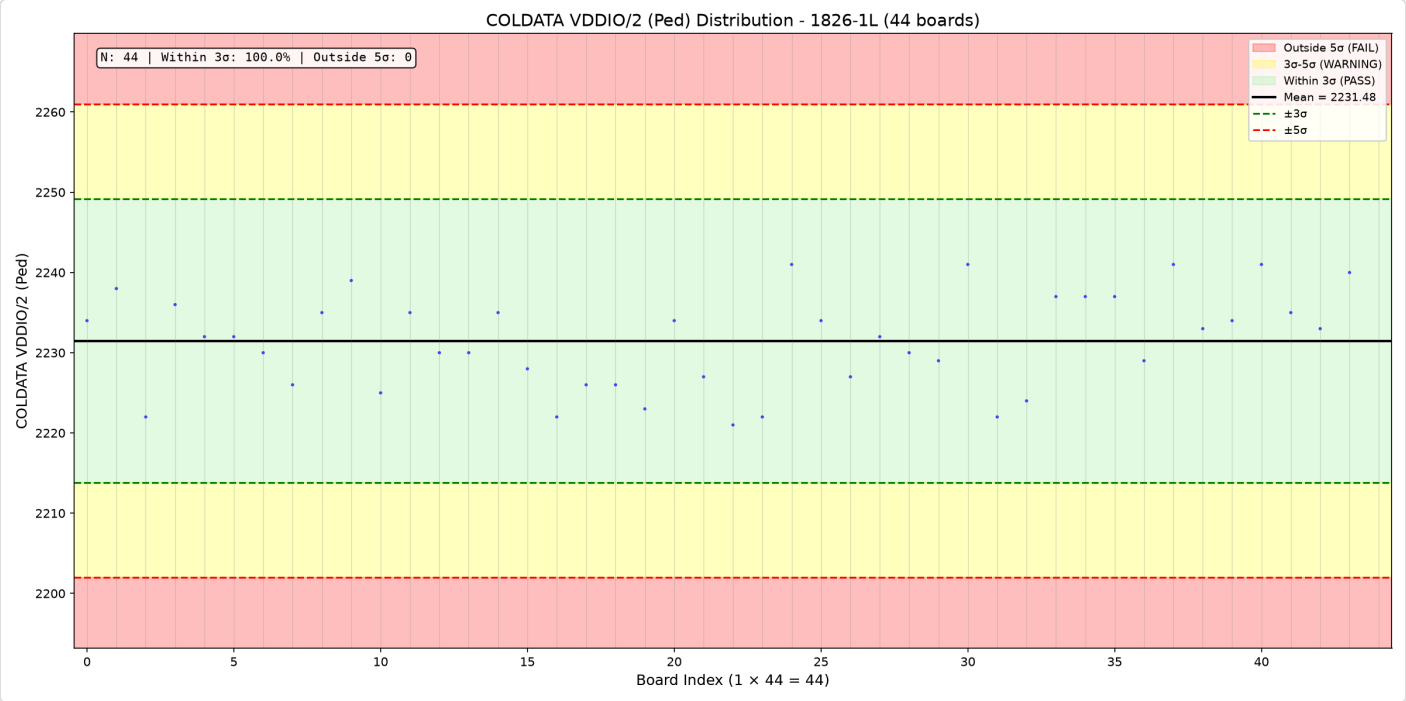
Statistic	Value
Mean	1185.16
Std	1.97
3σ Lower	1179.26
3σ Upper	1191.05
5σ Lower	1175.33
5σ Upper	1194.98



2.2 CDVDDIO_HALF_ped

Description: COLDATA I/O supply voltage (divided by 2 for ADC range). Powers the digital I/O of COLDATA.

Statistic	Value
Mean	2231.48
Std	5.89
3σ Lower	2213.79
3σ Upper	2249.16
5σ Lower	2202.00
5σ Upper	2260.95

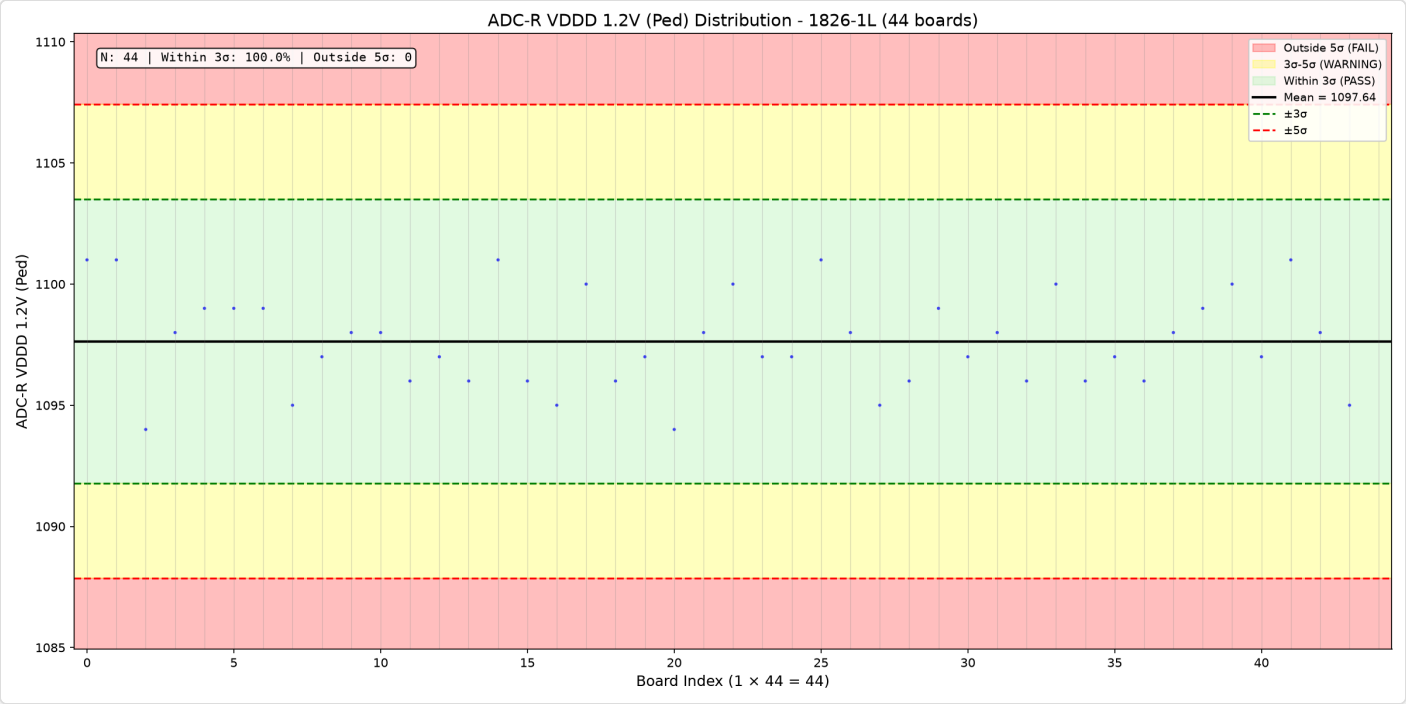


2.3 ADCRVDDD1P2_ped

Description: Right ColdADC digital 1.2V supply voltage. Powers the digital logic of the right-side ColdADC.

Statistic	Value
Mean	1097.64
Std	1.96
3σ Lower	1091.77

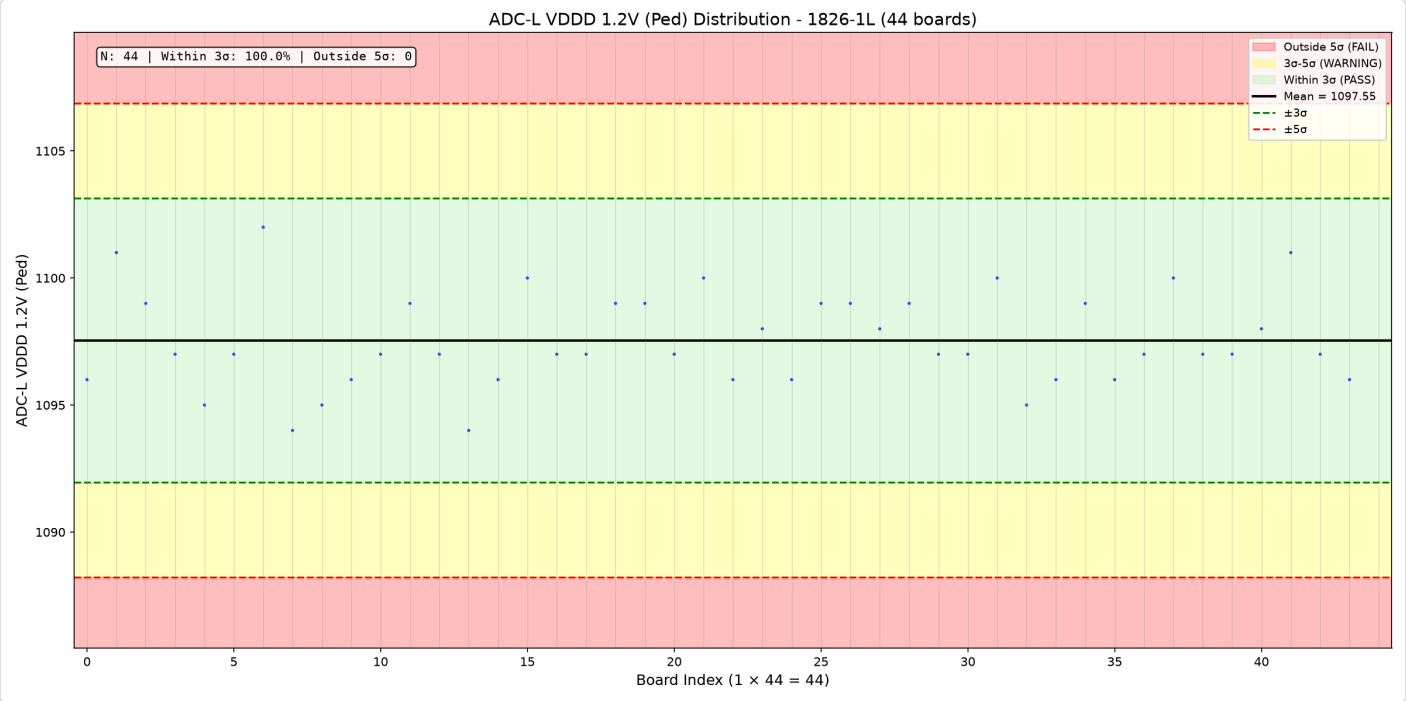
Statistic	Value
3σ Upper	1103.50
5σ Lower	1087.86
5σ Upper	1107.41



2.4 ADCLVDDD1P2_ped

Description: Left ColdADC digital 1.2V supply voltage. Powers the digital logic of the left-side ColdADC.

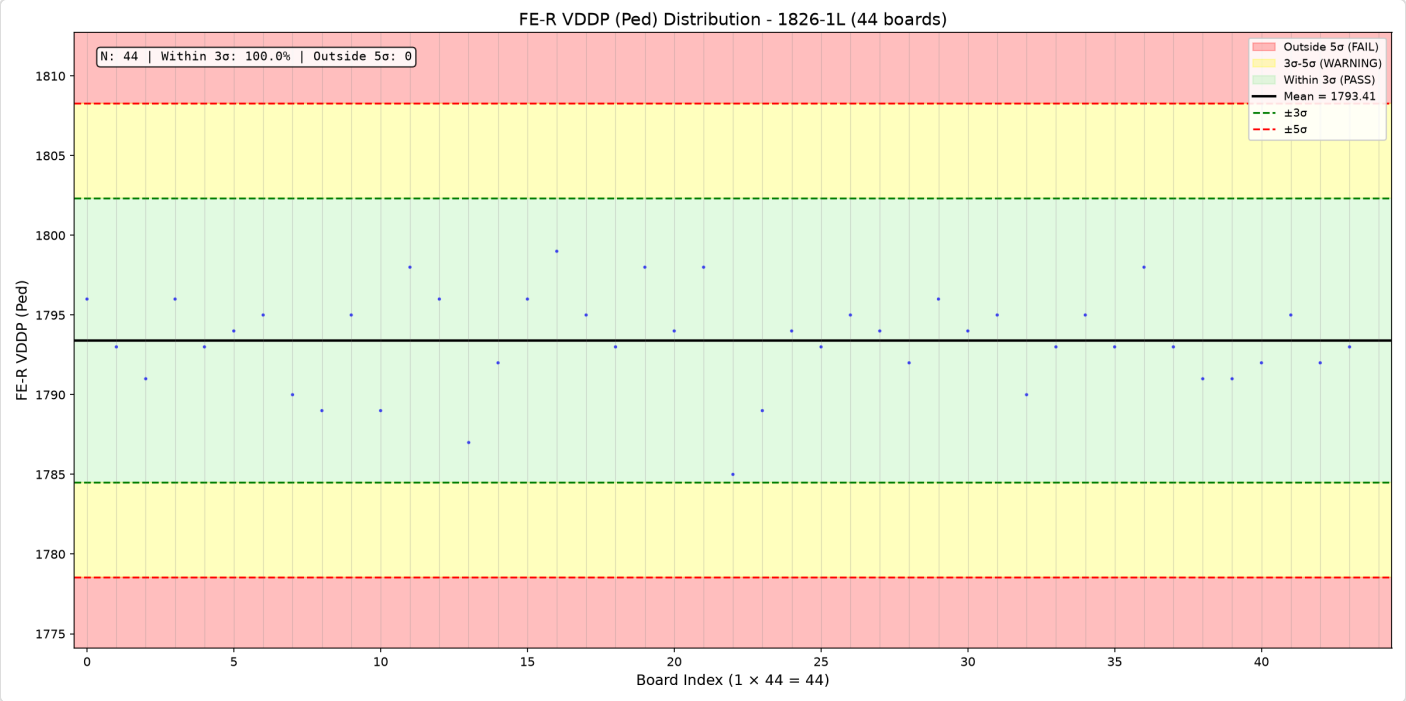
Statistic	Value
Mean	1097.55
Std	1.86
3σ Lower	1091.95
3σ Upper	1103.14
5σ Lower	1088.22
5σ Upper	1106.87



2.5 FERVDDP_ped

Description: Right LArASIC (FE) positive supply voltage during pedestal check. Powers the right-side front-end ASIC.

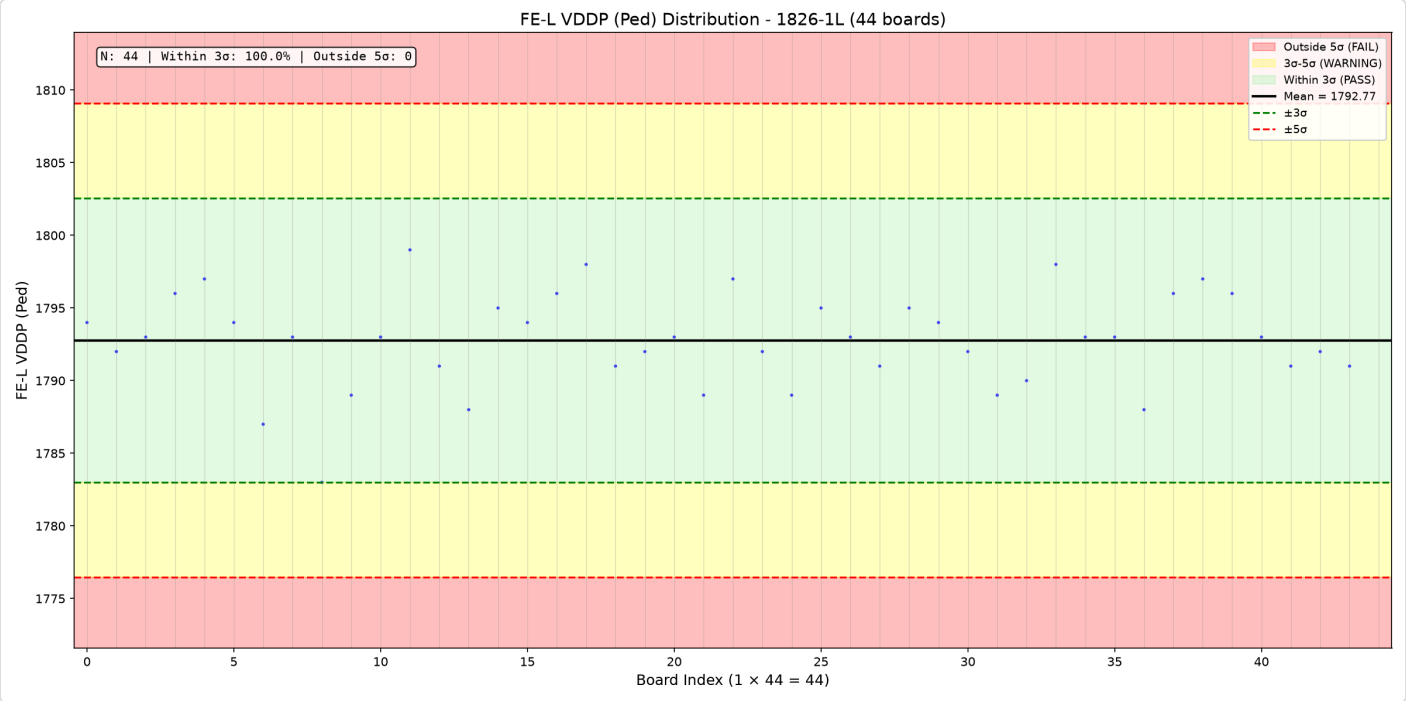
Statistic	Value
Mean	1793.41
Std	2.97
3σ Lower	1784.49
3σ Upper	1802.33
5σ Lower	1778.55
5σ Upper	1808.27



2.6 FELVDDP_ped

Description: Left LArASIC (FE) positive supply voltage during pedestal check.

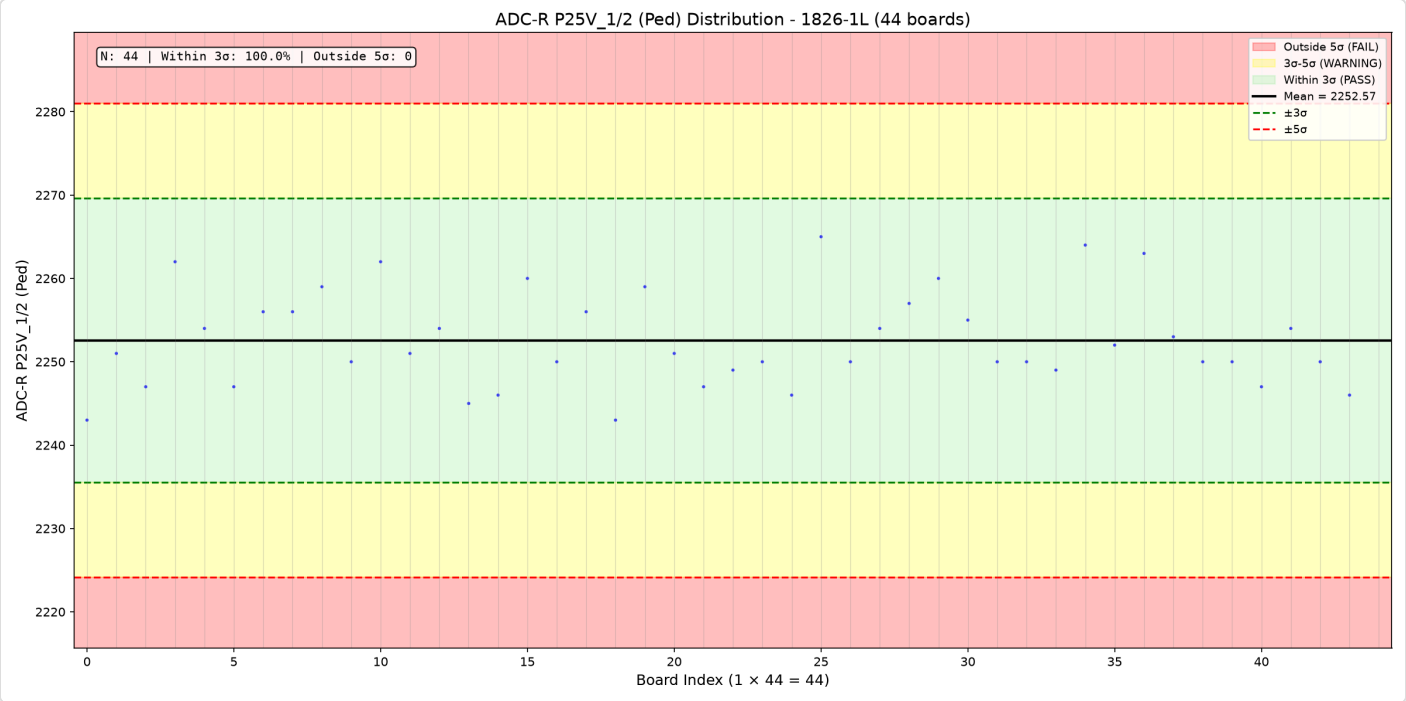
Statistic	Value
Mean	1792.77
Std	3.26
3σ Lower	1782.99
3σ Upper	1802.55
5σ Lower	1776.47
5σ Upper	1809.07



2.7 ADCRP25V_1_HALF_ped

Description: Right ColdADC 2.5V supply #1 (divided by 2). Analog reference voltage for right ADC.

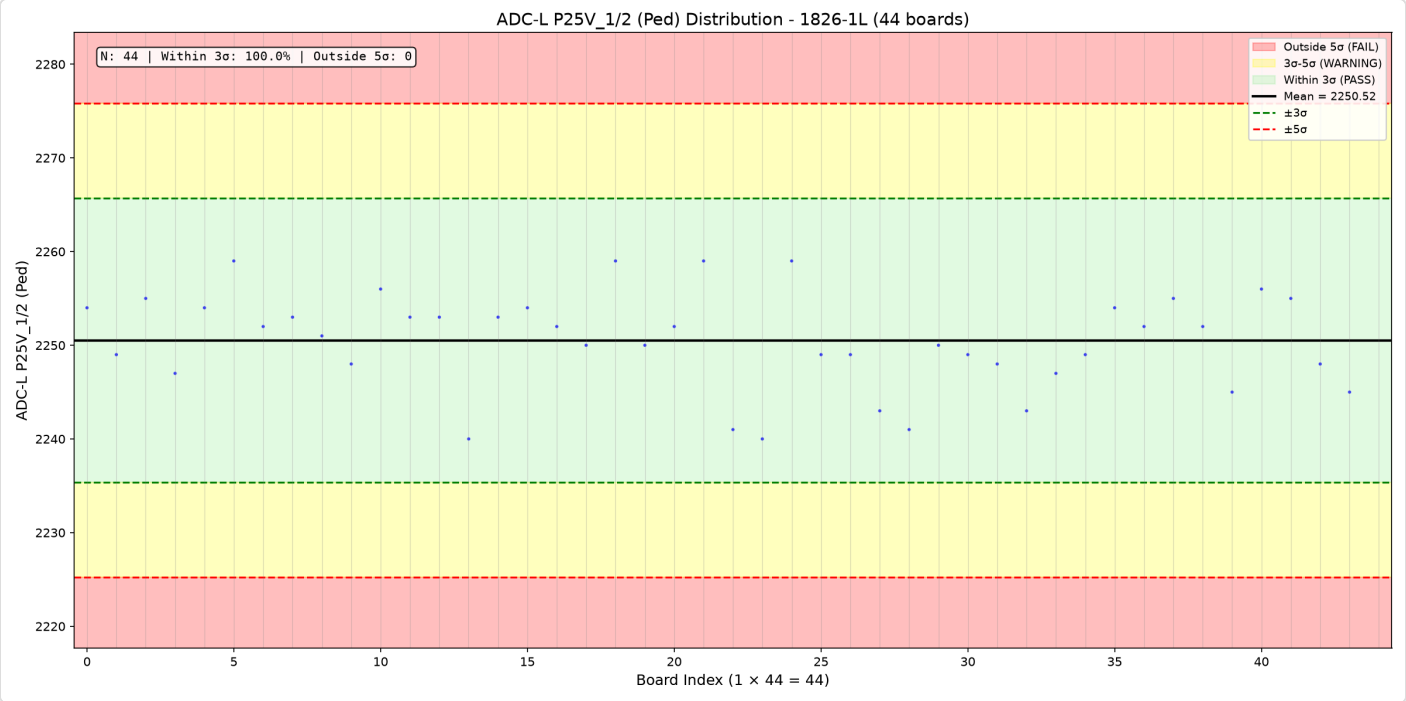
Statistic	Value
Mean	2252.57
Std	5.68
3σ Lower	2235.52
3σ Upper	2269.62
5σ Lower	2224.16
5σ Upper	2280.98



2.8 ADCLP25V_1_HALF_ped

Description: Left ColdADC 2.5V supply #1 (divided by 2).

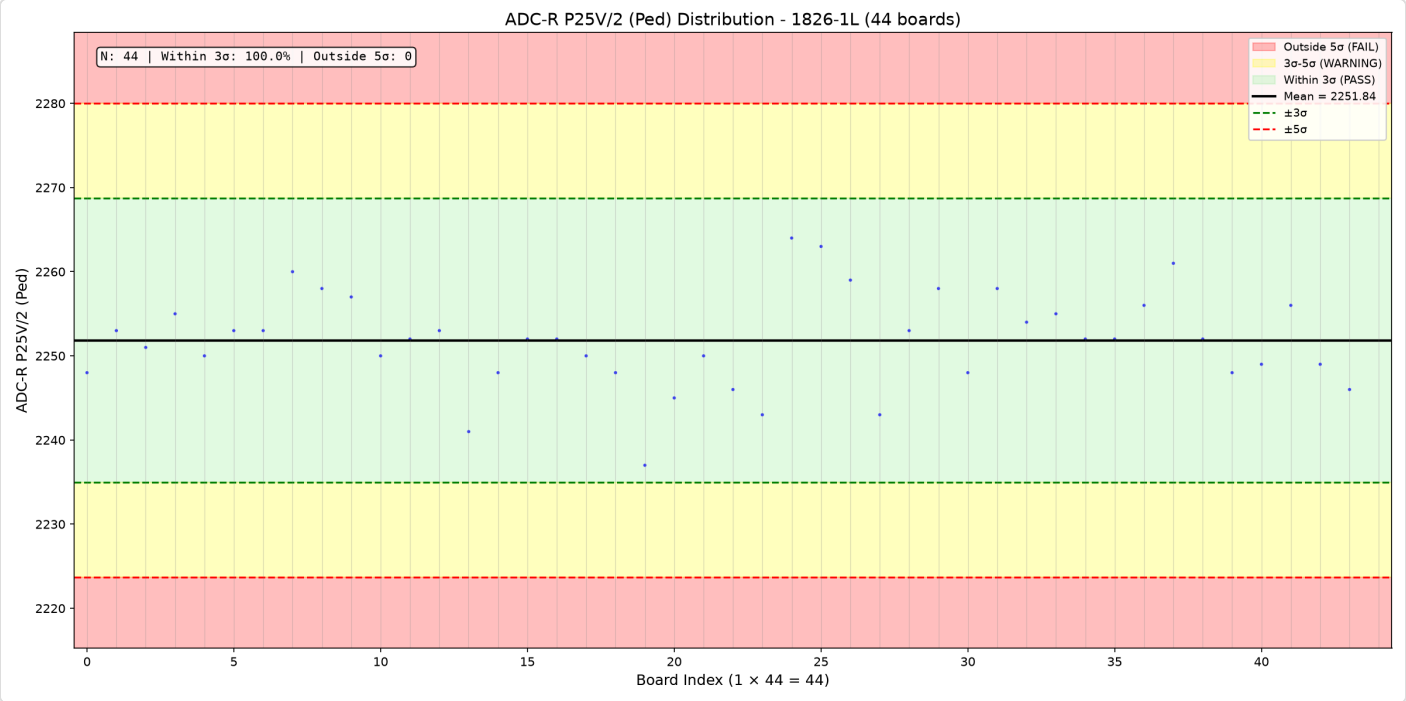
Statistic	Value
Mean	2250.52
Std	5.06
3σ Lower	2235.35
3σ Upper	2265.69
5σ Lower	2225.24
5σ Upper	2275.81



2.9 ADCRP25V_HALF_ped

Description: Right ColdADC 2.5V supply (divided by 2). Main 2.5V reference for right ADC.

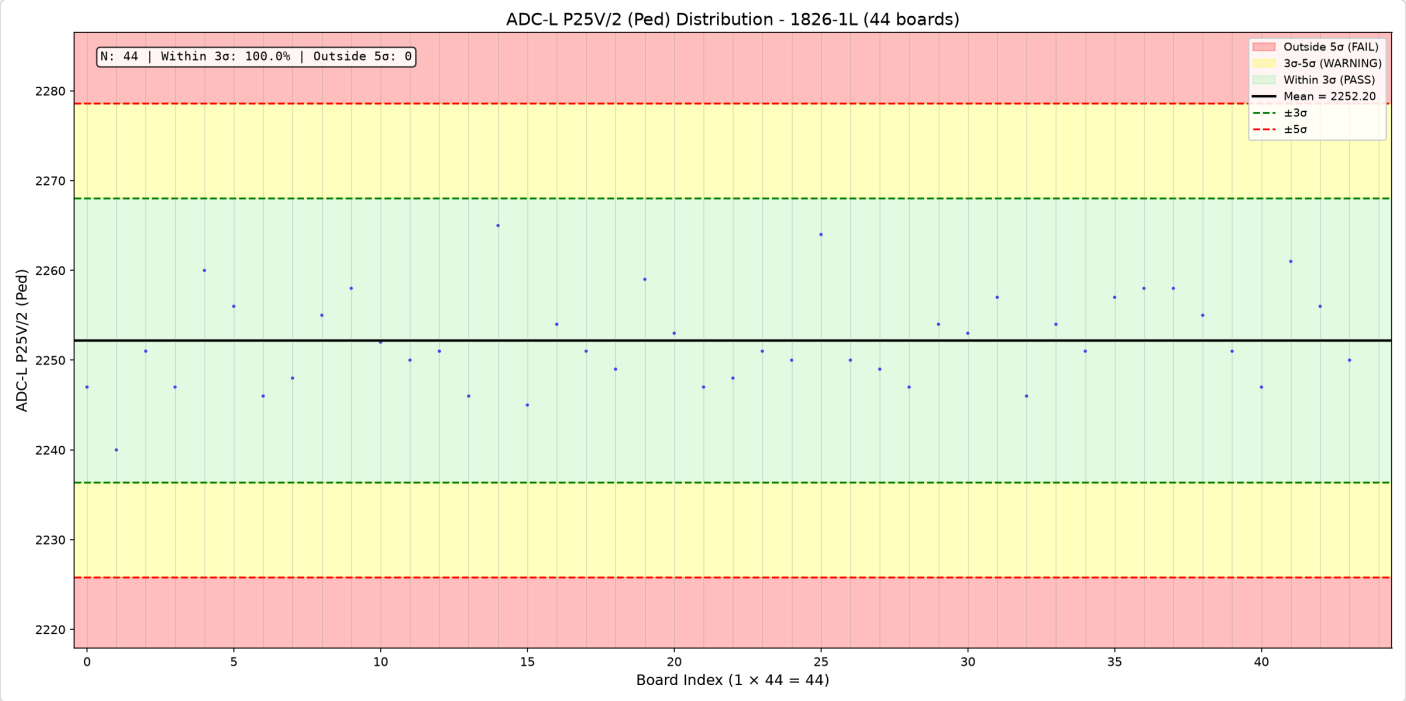
Statistic	Value
Mean	2251.84
Std	5.63
3σ Lower	2234.96
3σ Upper	2268.73
5σ Lower	2223.70
5σ Upper	2279.98



2.10 ADCLP25V_HALF_ped

Description: Left ColdADC 2.5V supply (divided by 2).

Statistic	Value
Mean	2252.20
Std	5.28
3σ Lower	2236.37
3σ Upper	2268.03
5σ Lower	2225.82
5σ Upper	2278.59

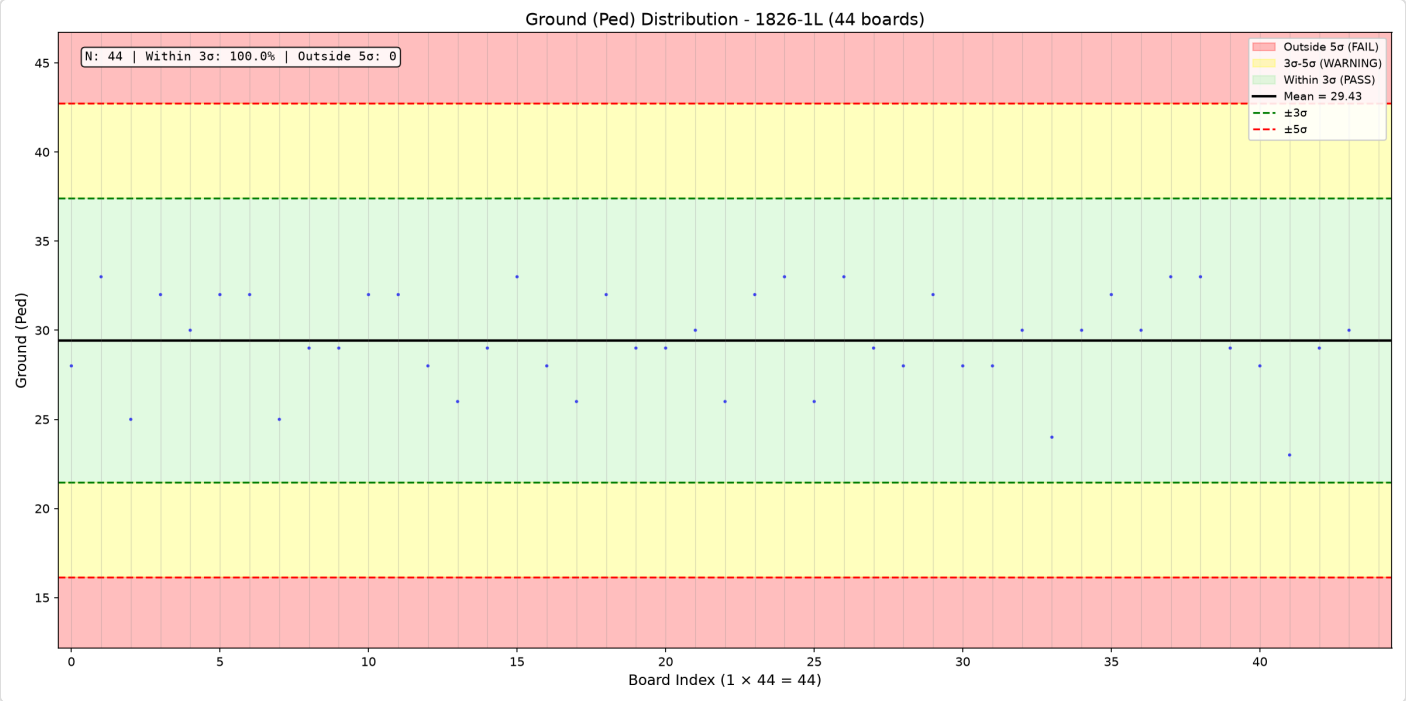


2.11 GND_ped

Description: Ground reference measurement during pedestal check. Should be near zero.

Statistic	Value
Mean	29.43
Std	2.66
3σ Lower	21.46
3σ Upper	37.41
5σ Lower	16.14
5σ Upper	42.72

Interpretation: Non-zero value represents ADC offset. Should be consistent across boards.



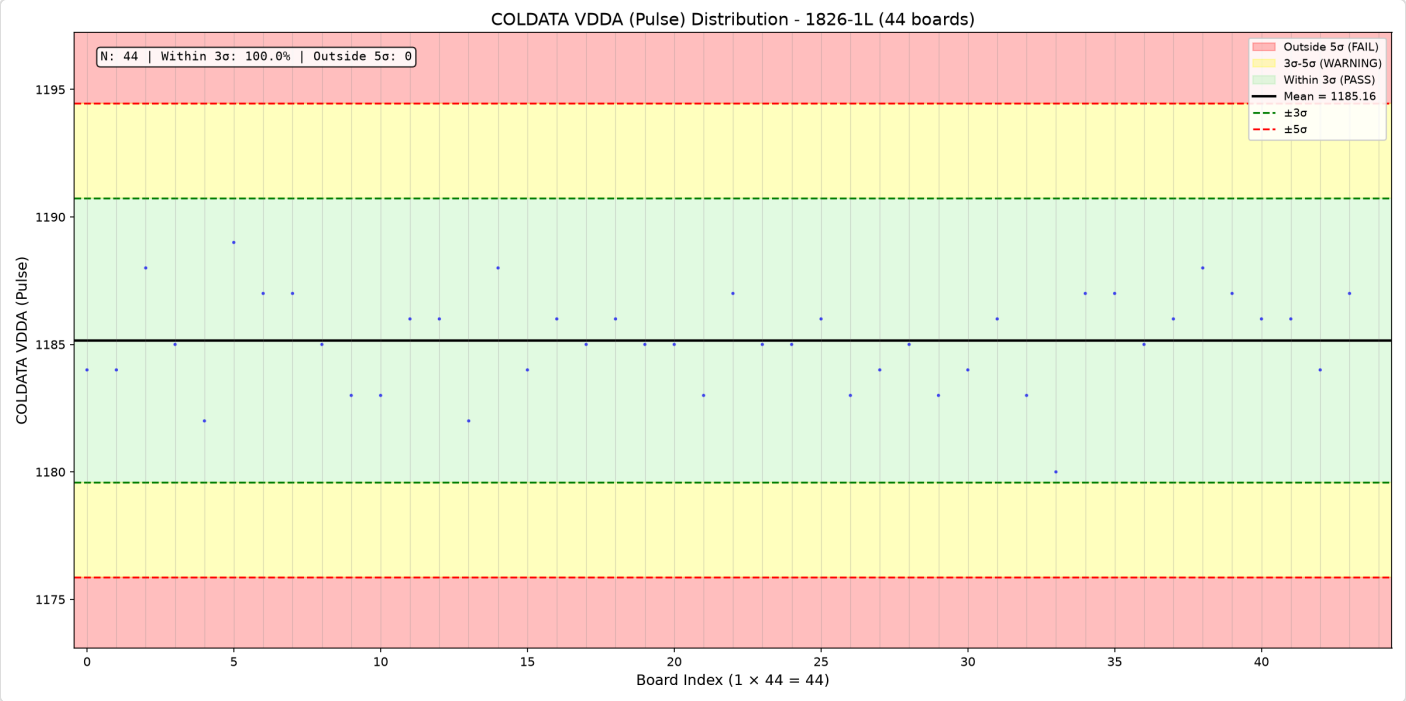
3. LDO Voltage Parameters (Pulse Configuration)

Same LDO measurements during pulse response check. Different operating conditions may show slight voltage shifts.

3.1 CDVDDA_pulse

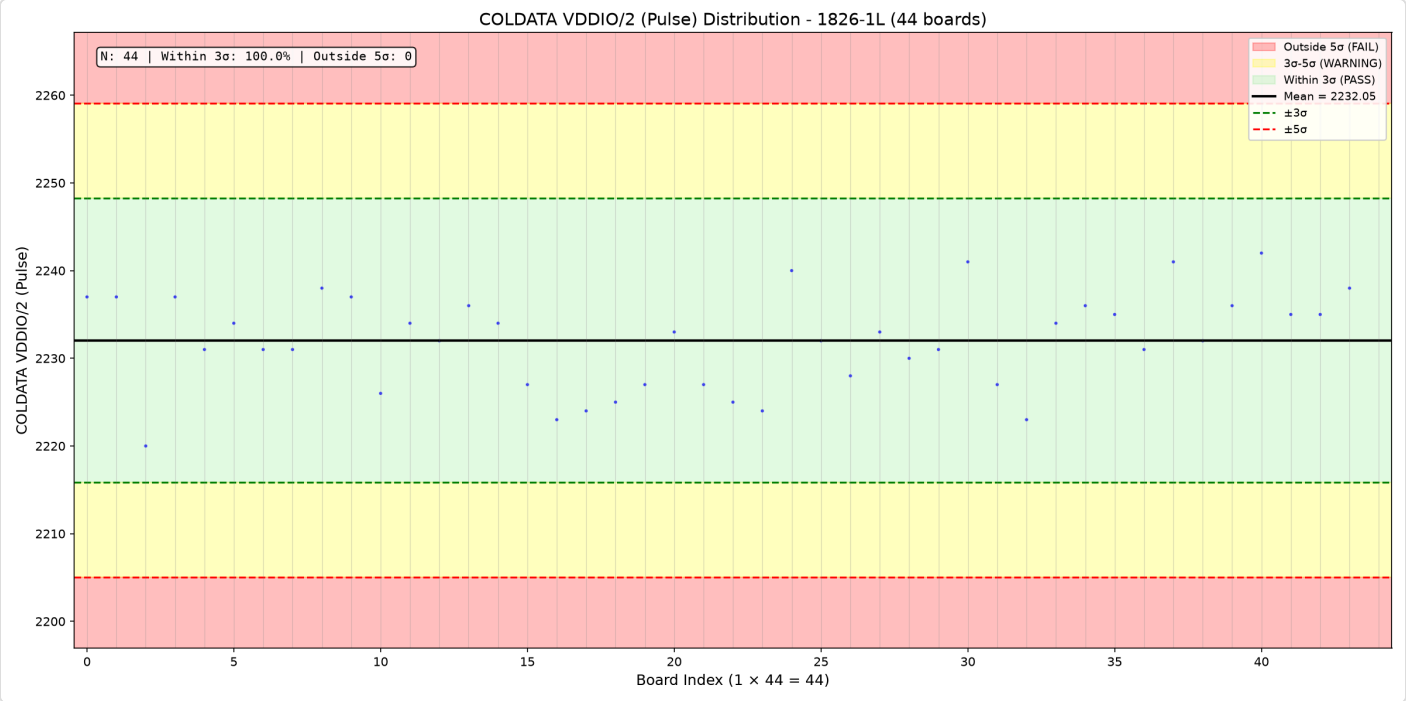
Description: COLDATA analog supply voltage during pulse check.

Statistic	Value
Mean	1185.16
Std	1.86
5σ Lower	1175.87
5σ Upper	1194.45



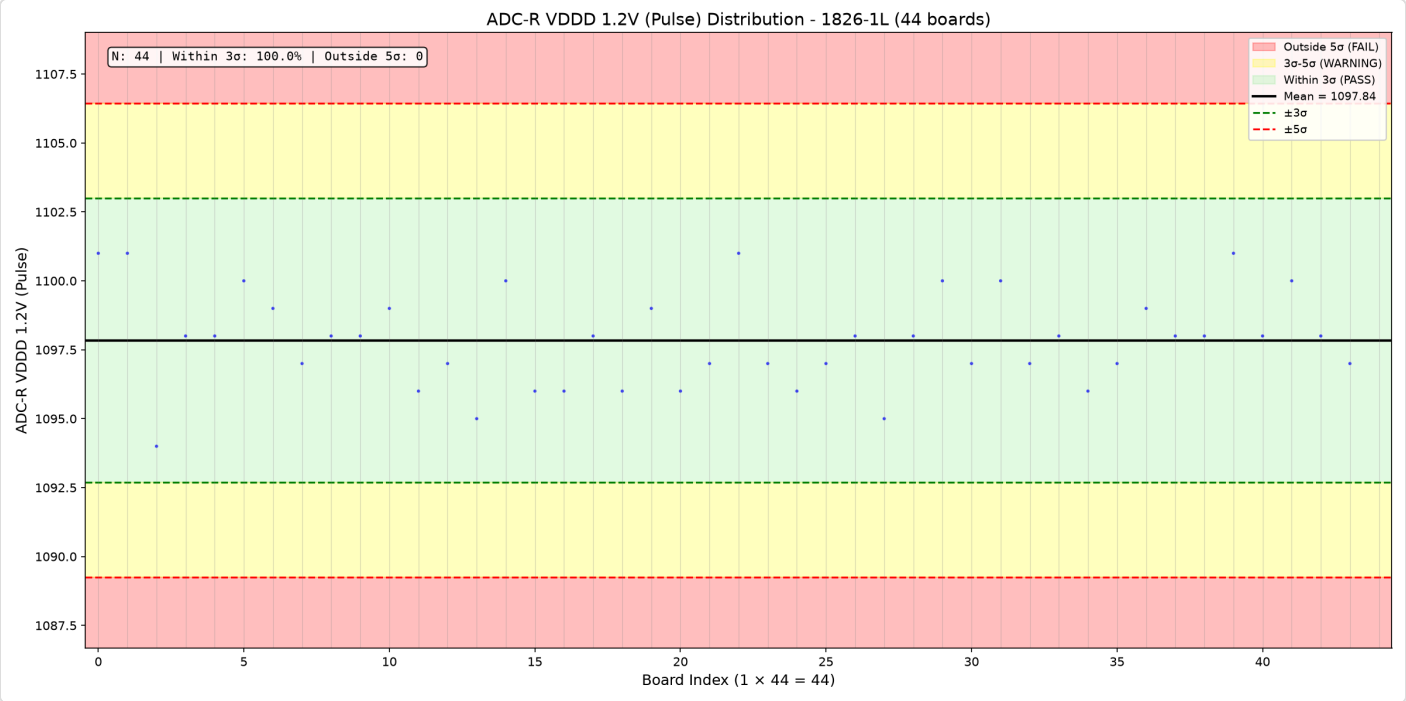
3.2 CDVDDIO_HALF_pulse

Statistic	Value
Mean	2232.05
Std	5.40
5σ Lower	2205.04
5σ Upper	2259.05



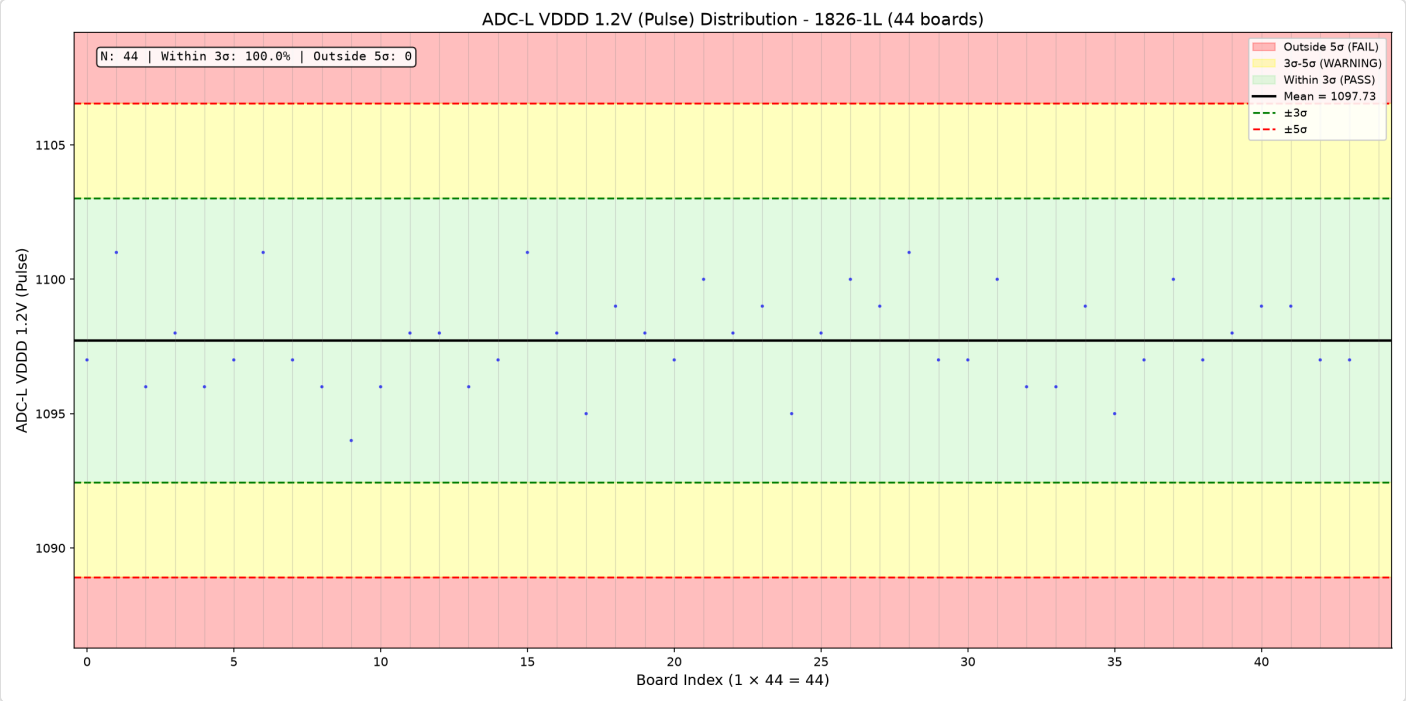
3.3 ADCRVDDD1P2_pulse

Statistic	Value
Mean	1097.84
Std	1.72
5σ Lower	1089.25
5σ Upper	1106.43



3.4 ADCLVDDD1P2_pulse

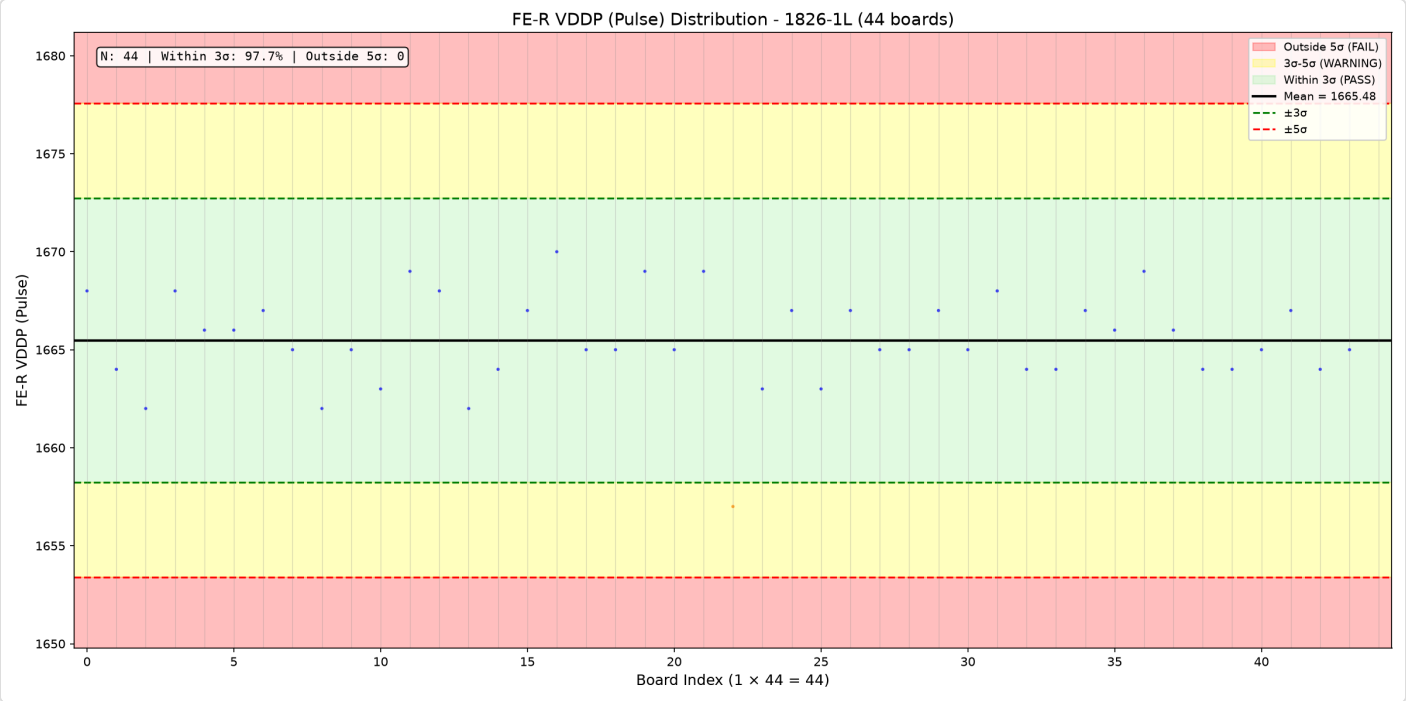
Statistic	Value
Mean	1097.73
Std	1.76
5σ Lower	1088.91
5σ Upper	1106.54



3.5 FERVDDP_pulse

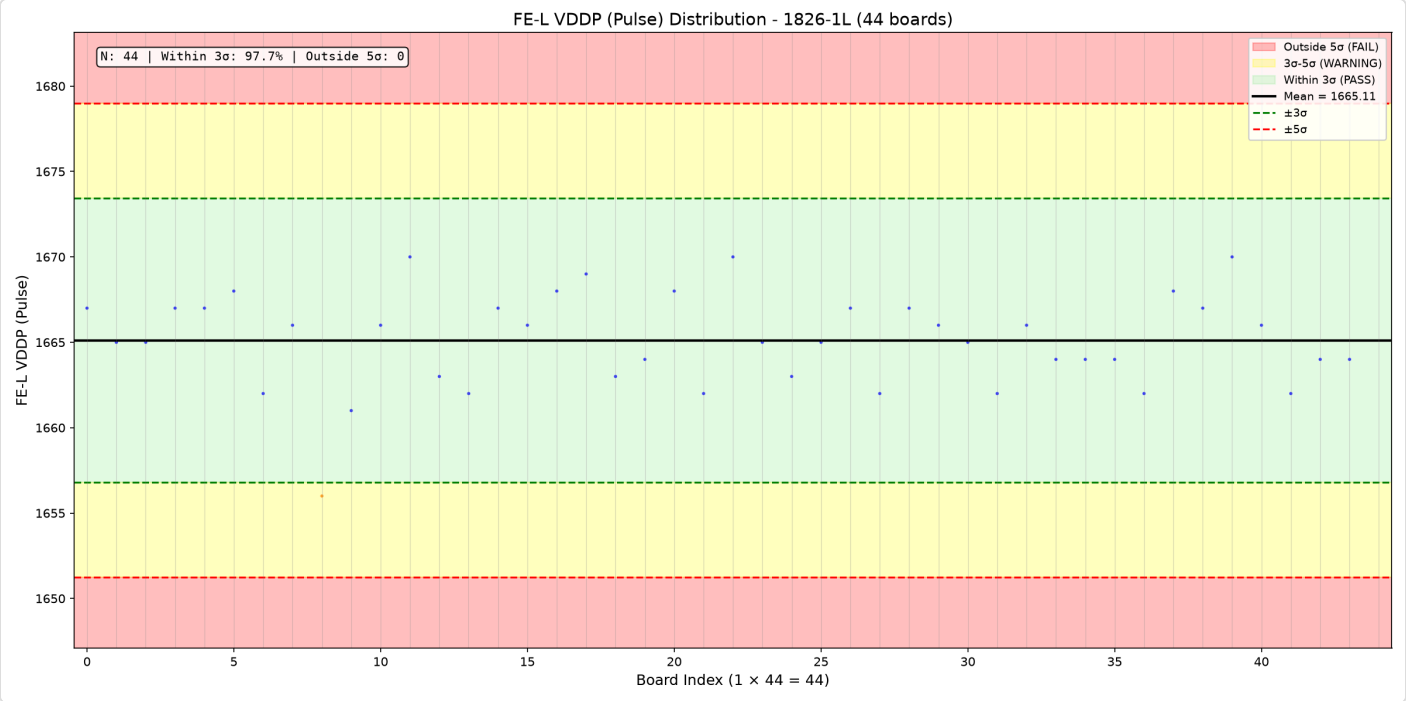
Statistic	Value
Mean	1665.48
Std	2.42
5σ Lower	1653.39
5σ Upper	1677.56

Note: FERVDDP is lower during pulse than pedestal due to increased current draw.



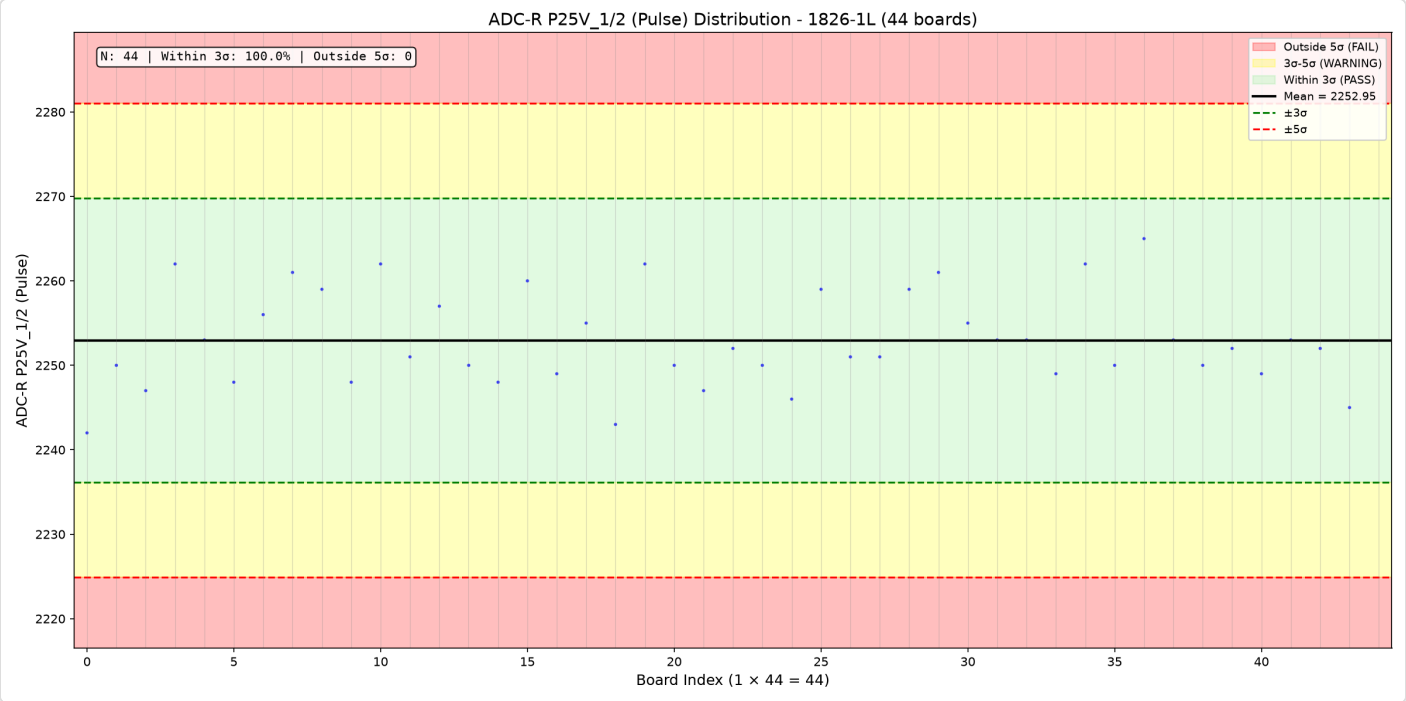
3.6 FELVDDP_pulse

Statistic	Value
Mean	1665.11
Std	2.77
5σ Lower	1651.25
5σ Upper	1678.98



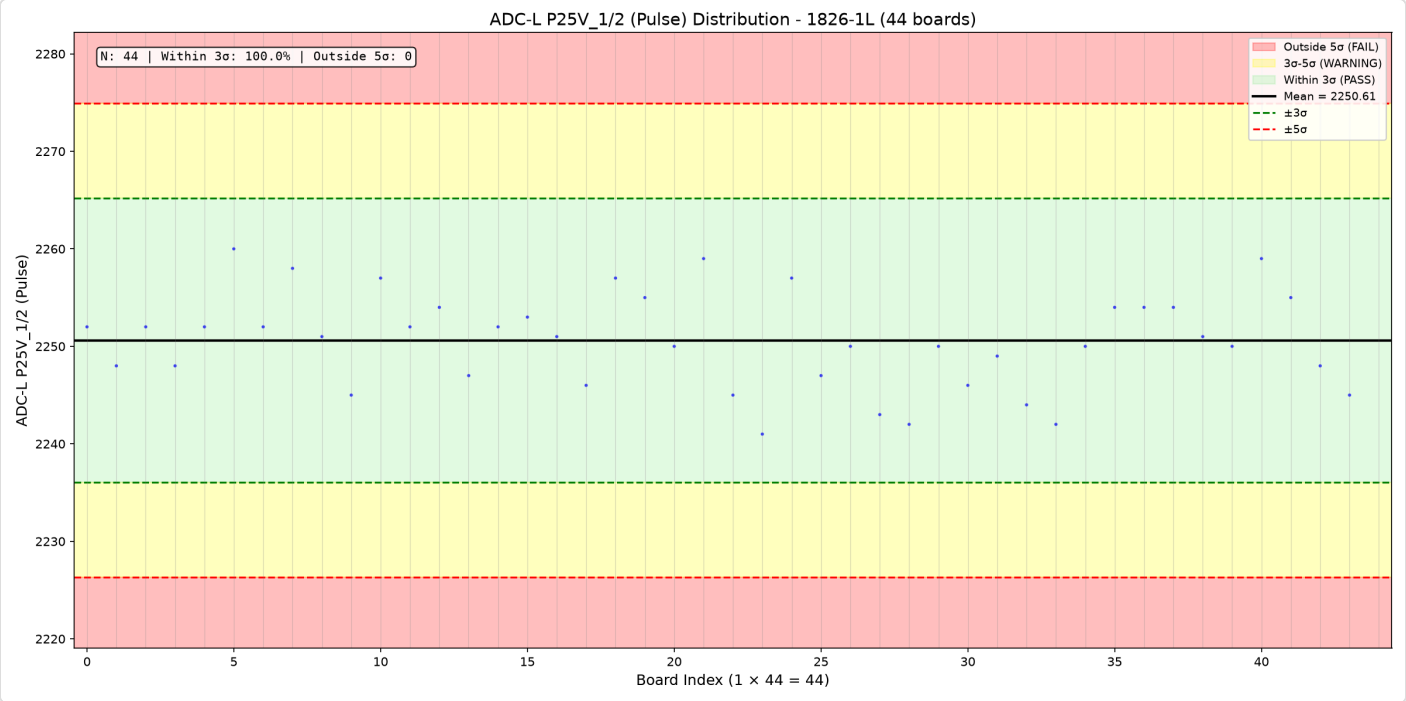
3.7 ADCRP25V_1_HALF_pulse

Statistic	Value
Mean	2252.95
Std	5.61
5σ Lower	2224.91
5σ Upper	2281.00



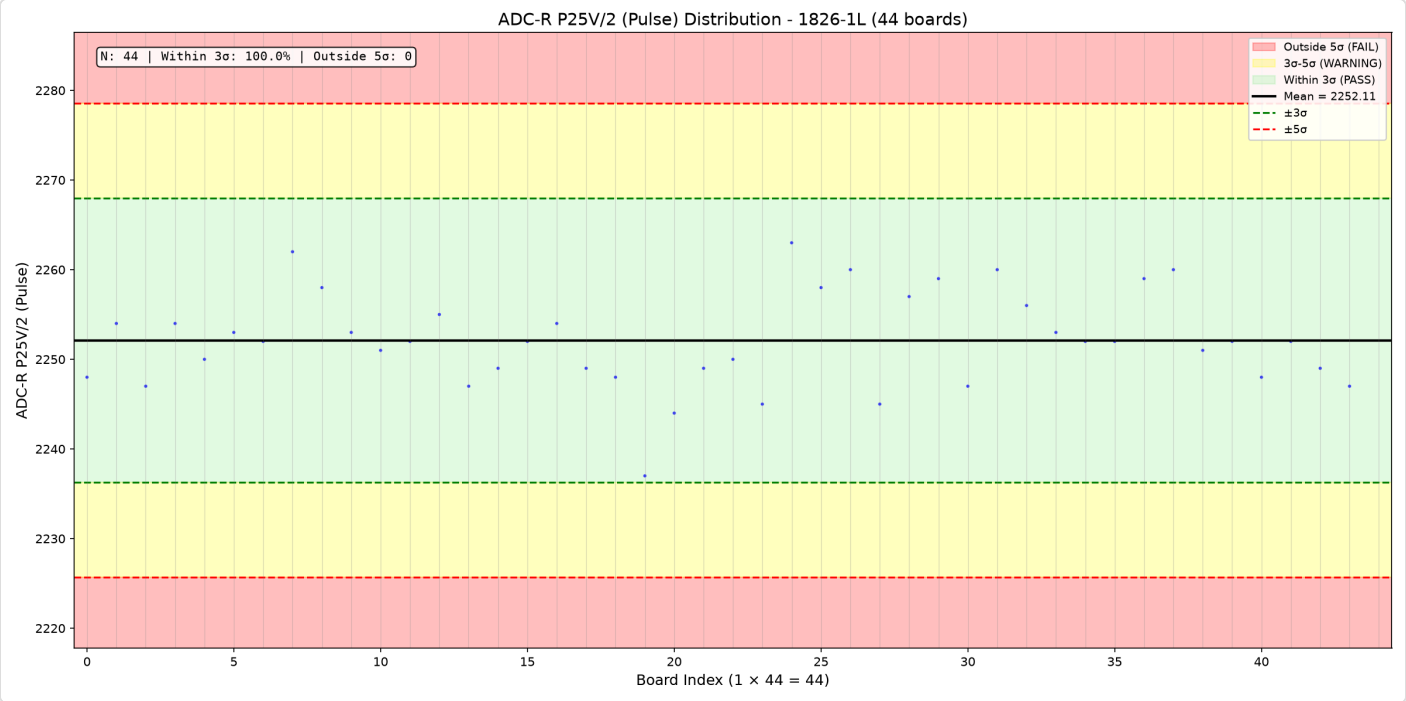
3.8 ADCLP25V_1_HALF_pulse

Statistic	Value
Mean	2250.61
Std	4.86
5σ Lower	2226.32
5σ Upper	2274.90



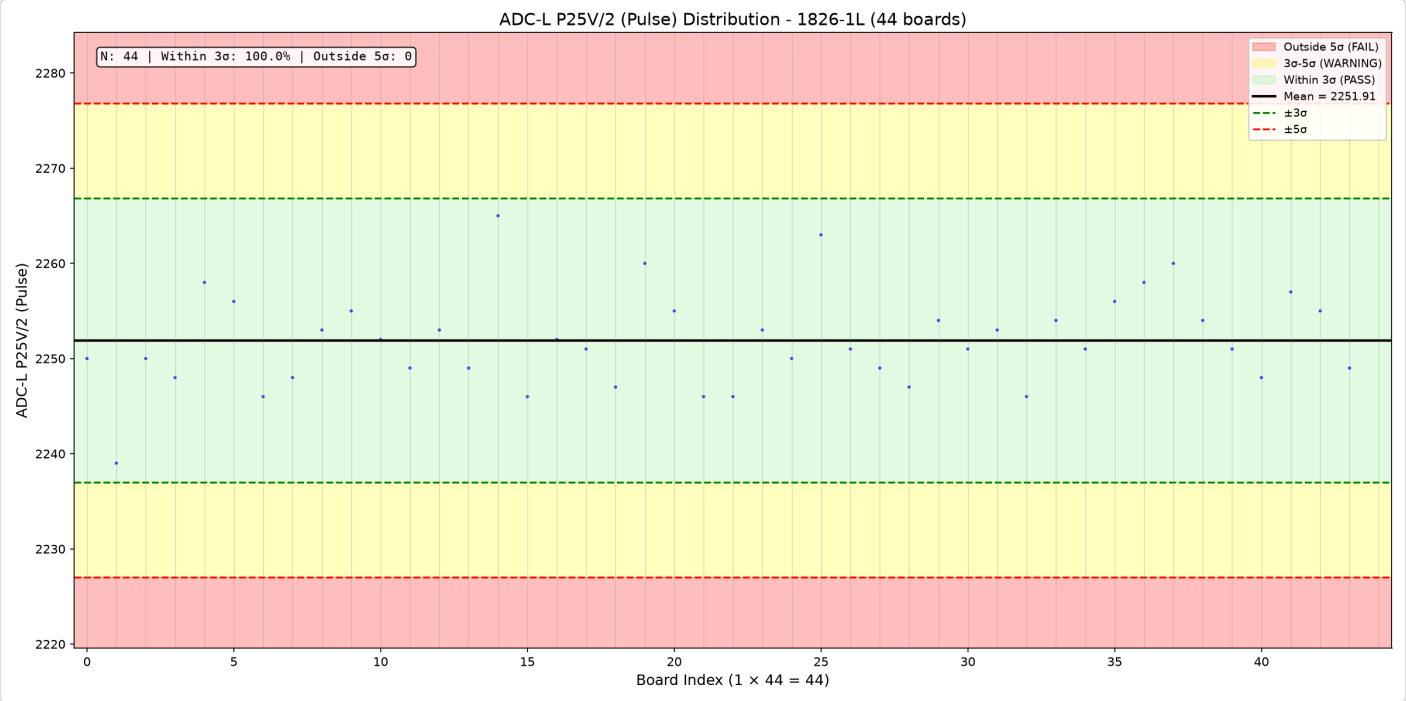
3.9 ADCRP25V_HALF_pulse

Statistic	Value
Mean	2252.11
Std	5.28
5σ Lower	2225.69
5σ Upper	2278.53



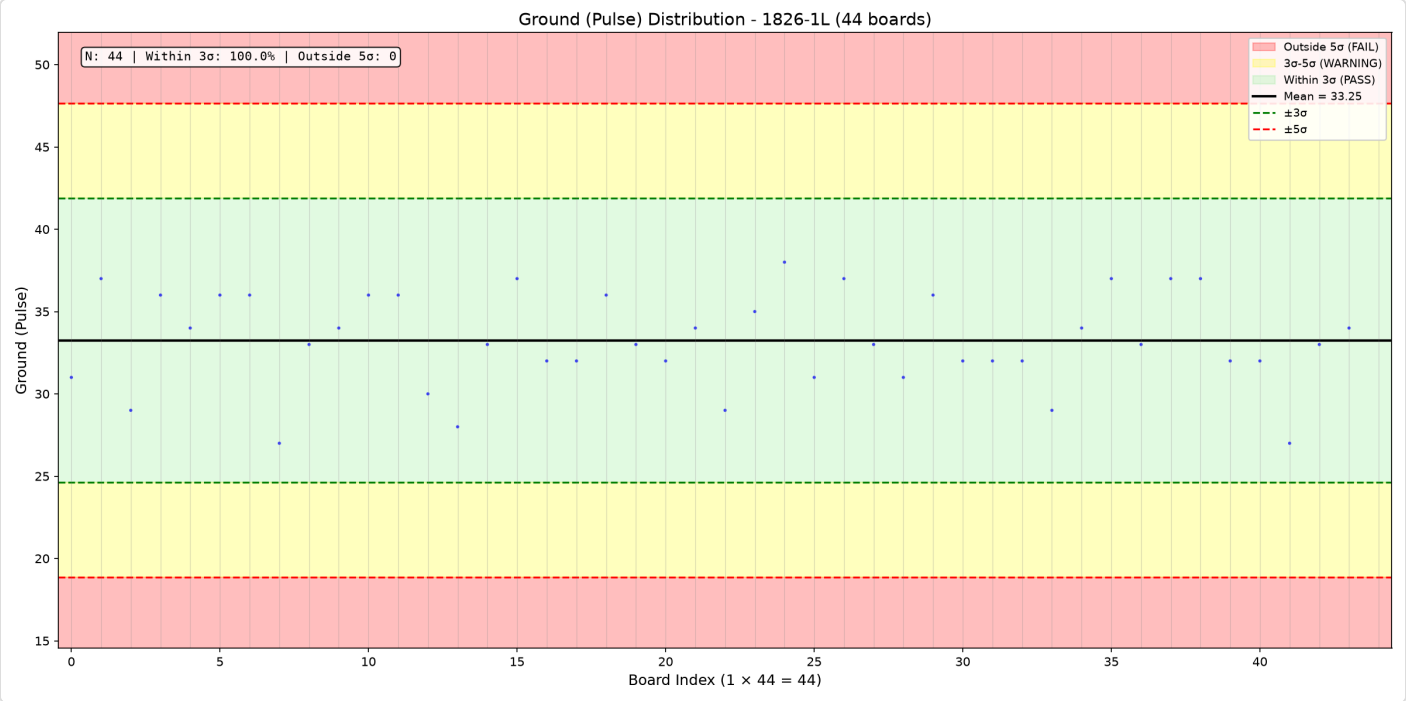
3.10 ADCLP25V_HALF_pulse

Statistic	Value
Mean	2251.91
Std	4.98
5σ Lower	2227.03
5σ Upper	2276.79



3.11 GND_pulse

Statistic	Value
Mean	33.25
Std	2.88
5σ Lower	18.86
5σ Upper	47.64



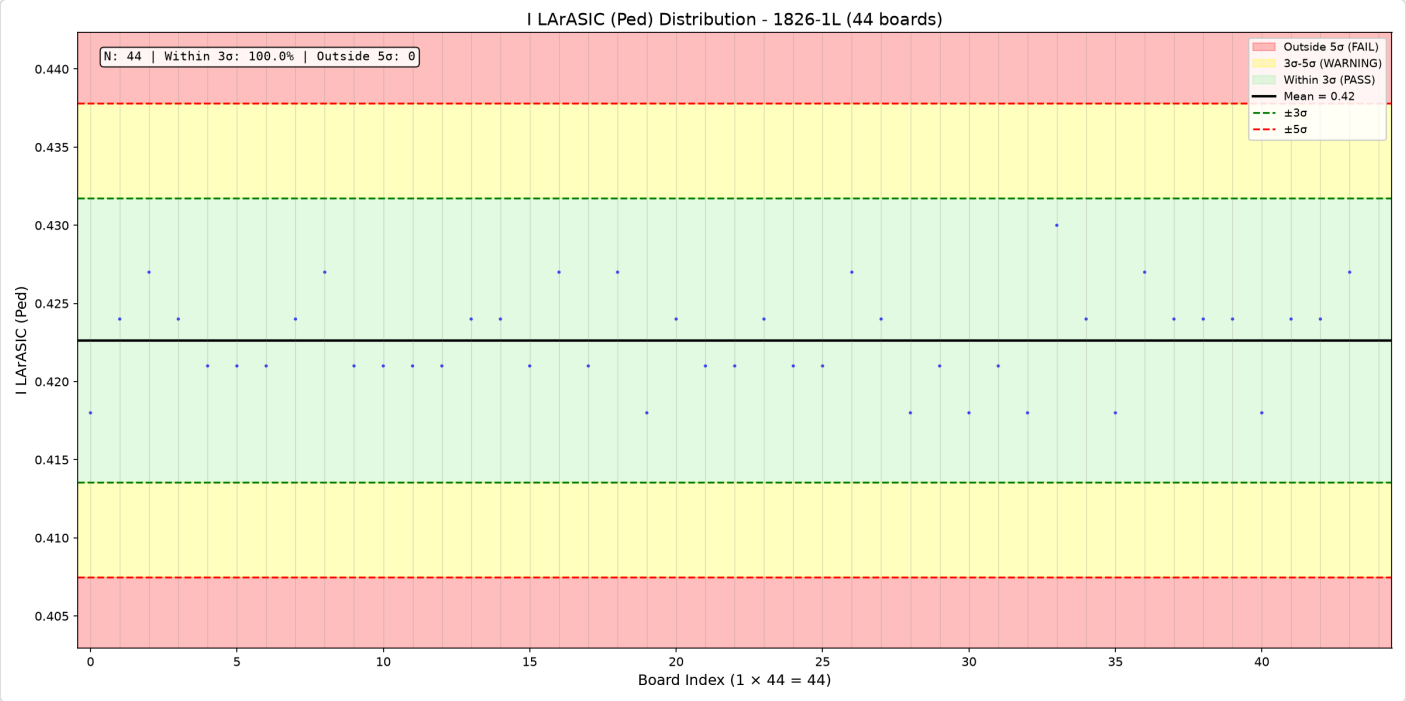
4. Current Measurements

Board-level current consumption for each power rail, measured at different operating configurations.

4.1 I_LArASIC (Pedestal)

Description: LArASIC front-end ASIC current during pedestal measurement. Lower power state.

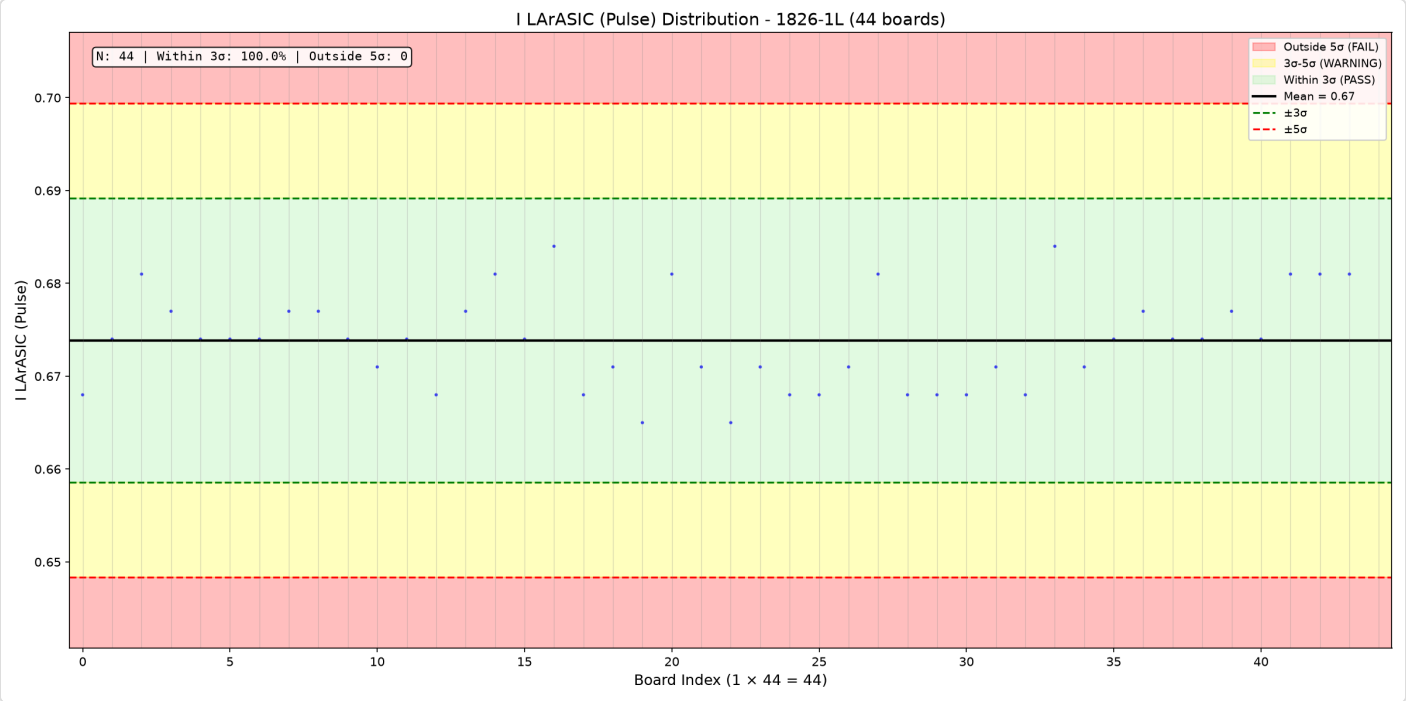
Statistic	Value (A)
Mean	0.4226
Std	0.0030
5σ Lower	0.4075
5σ Upper	0.4378



4.2 I_LArASIC (Pulse)

Description: LArASIC current during pulse response. Higher due to active pulsing.

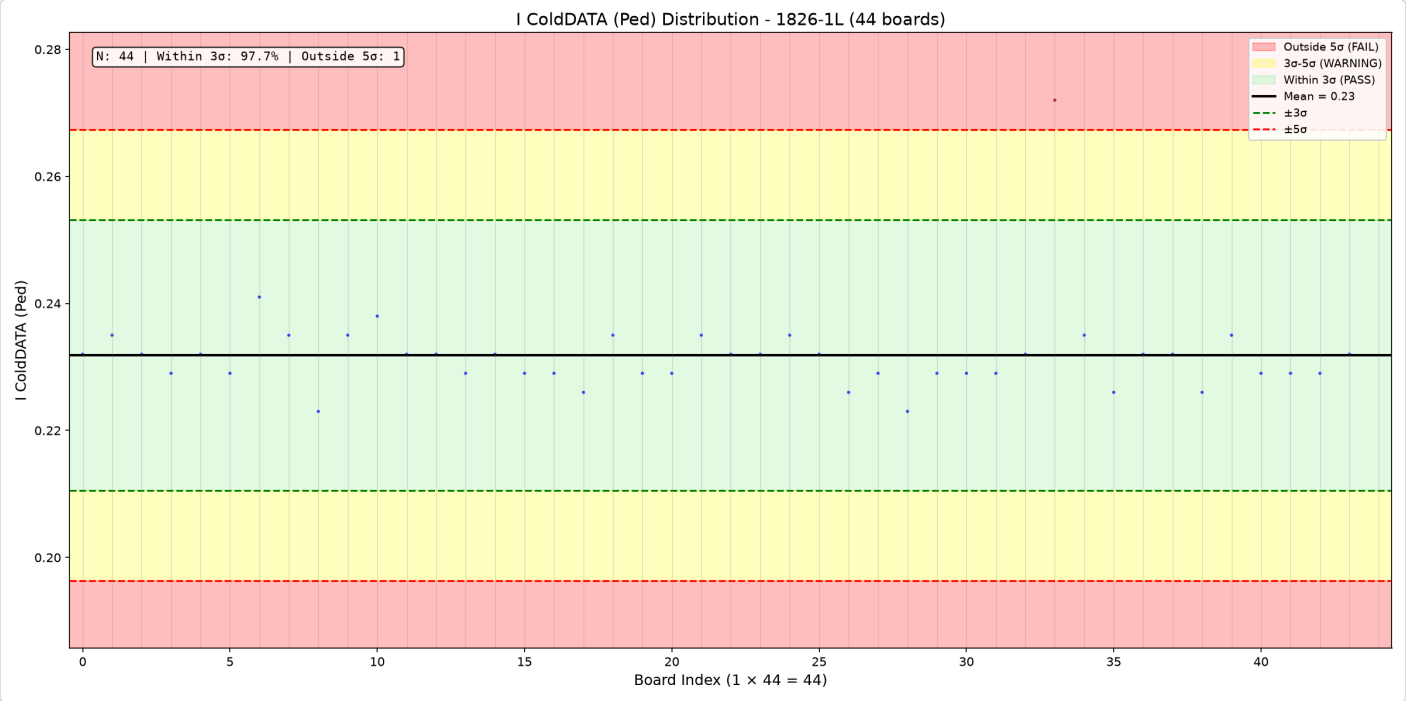
Statistic	Value (A)
Mean	0.6739
Std	0.0051
5σ Lower	0.6484
5σ Upper	0.6994



4.3 I_ColdDATA (Pedestal)

Description: COLDATA serializer ASIC current during pedestal check.

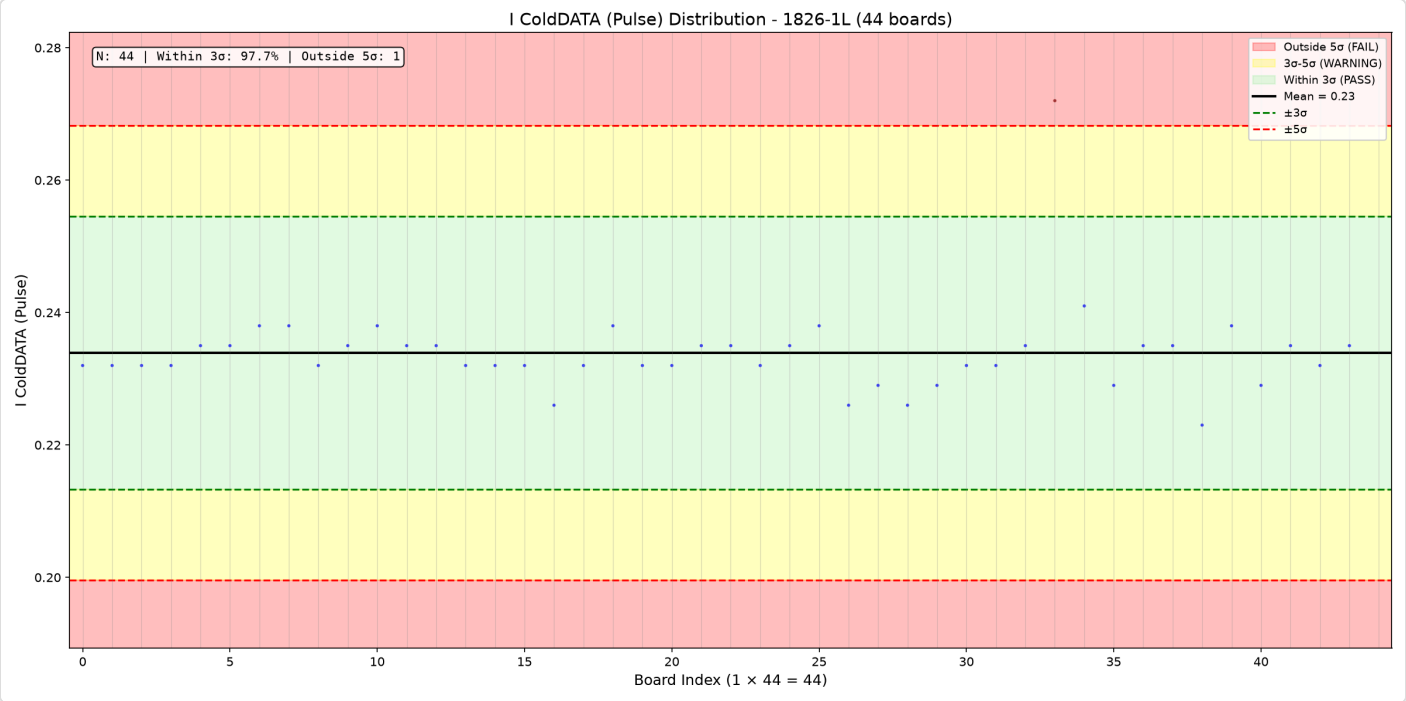
Statistic	Value (A)
Mean	0.2319
Std	0.0071
5σ Lower	0.1964
5σ Upper	0.2674



4.4 I_ColdDATA (Pulse)

Description: COLDATA current during pulse response.

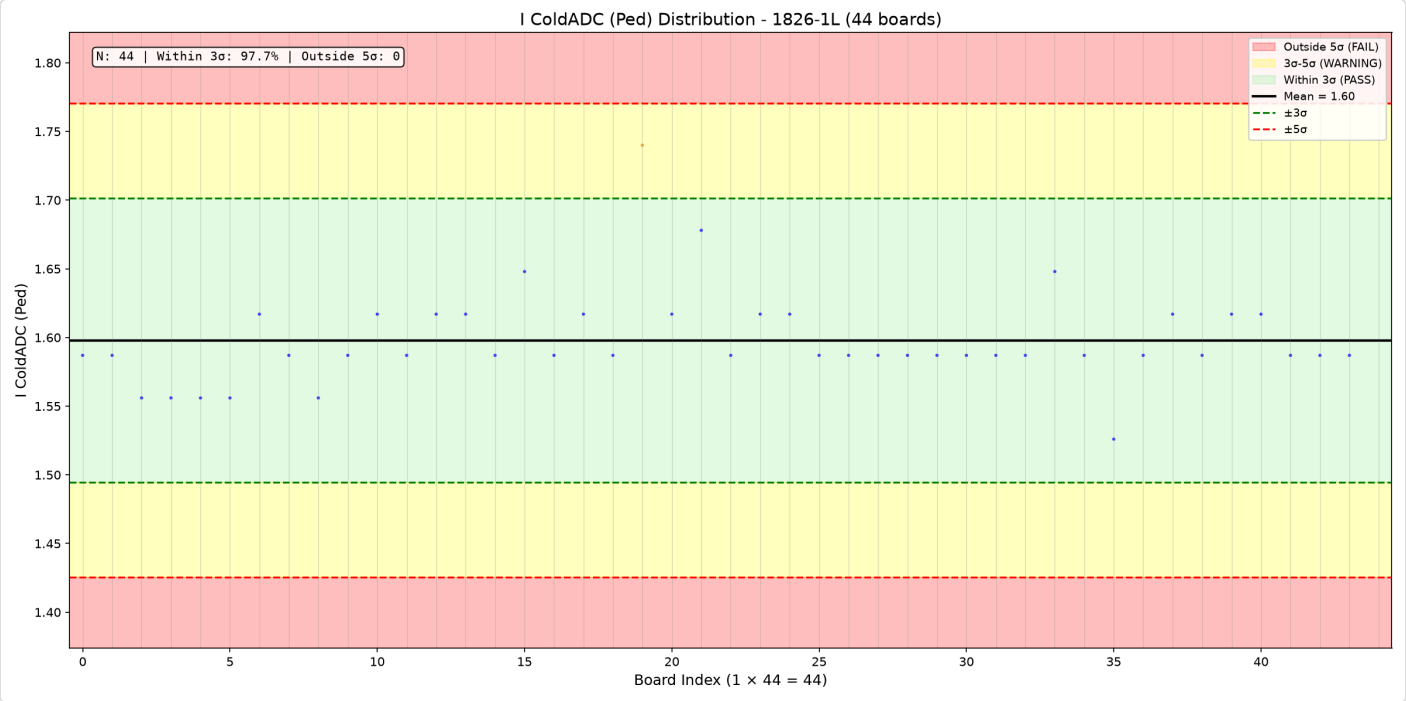
Statistic	Value (A)
Mean	0.2339
Std	0.0069
5σ Lower	0.1996
5σ Upper	0.2683



4.5 I_ColdADC (Pedestal)

Description: ColdADC digitizer current during pedestal check.

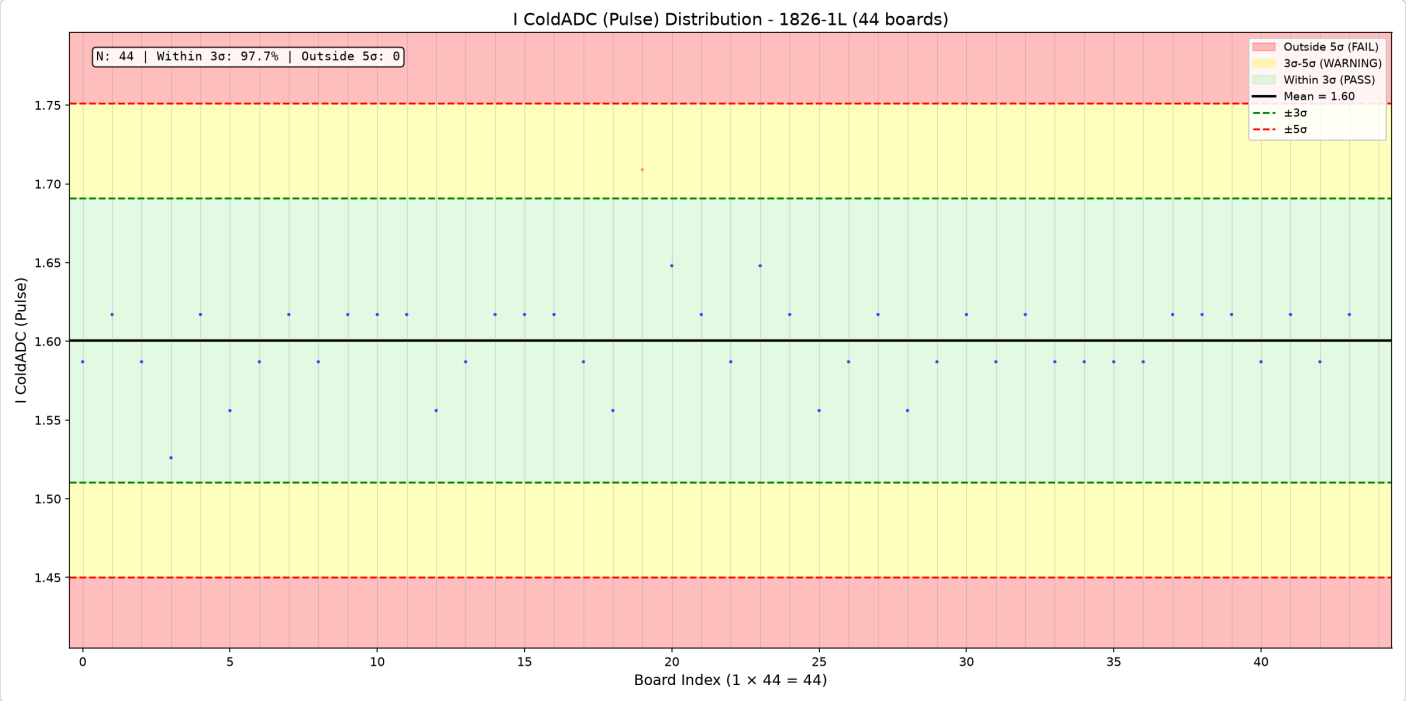
Statistic	Value (A)
Mean	1.5979
Std	0.0345
5σ Lower	1.4255
5σ Upper	1.7703



4.6 I_ColdADC (Pulse)

Description: ColdADC current during pulse response.

Statistic	Value (A)
Mean	1.6006
Std	0.0301
5σ Lower	1.4502
5σ Upper	1.7510

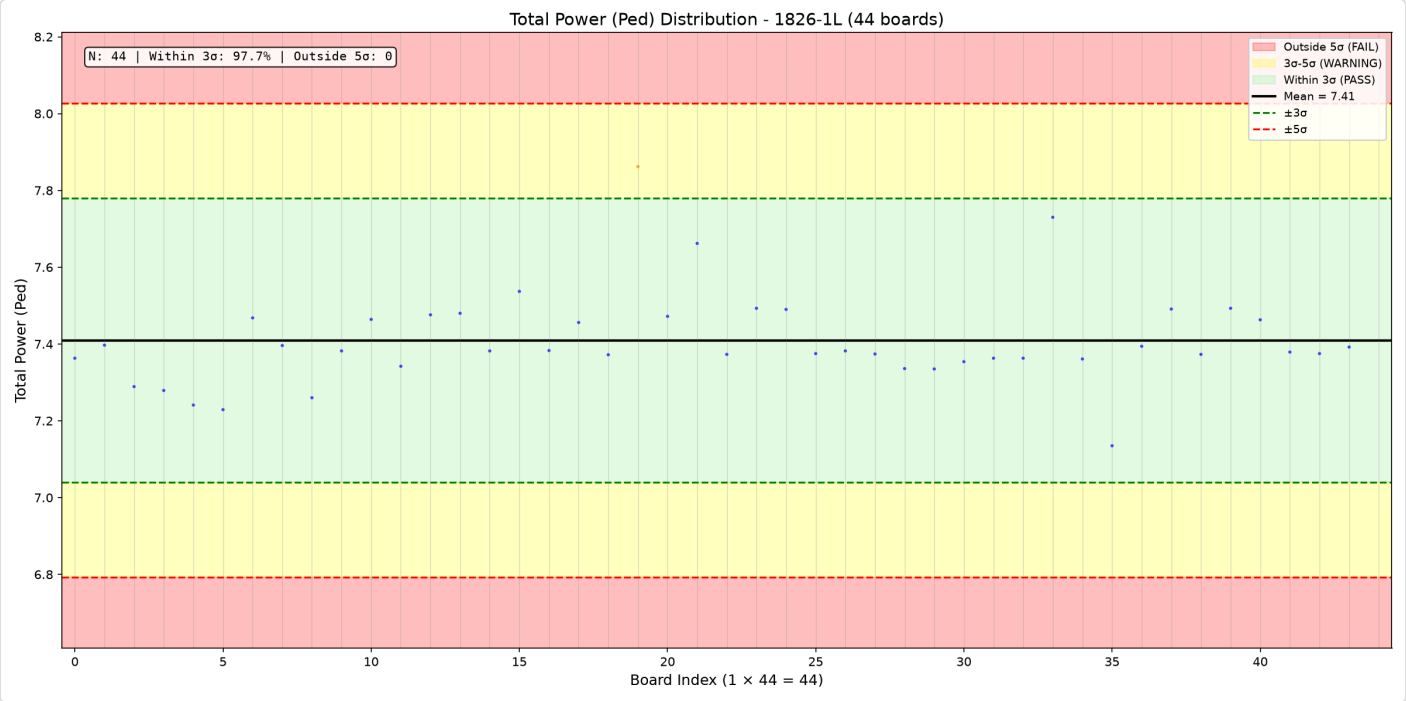


5. Power Measurements

5.1 Total Power (Pedestal)

Description: Total board power consumption during pedestal check configuration.

Statistic	Value (W)
Mean	7.4095
Std	0.1233
5σ Lower	6.7927
5σ Upper	8.0262

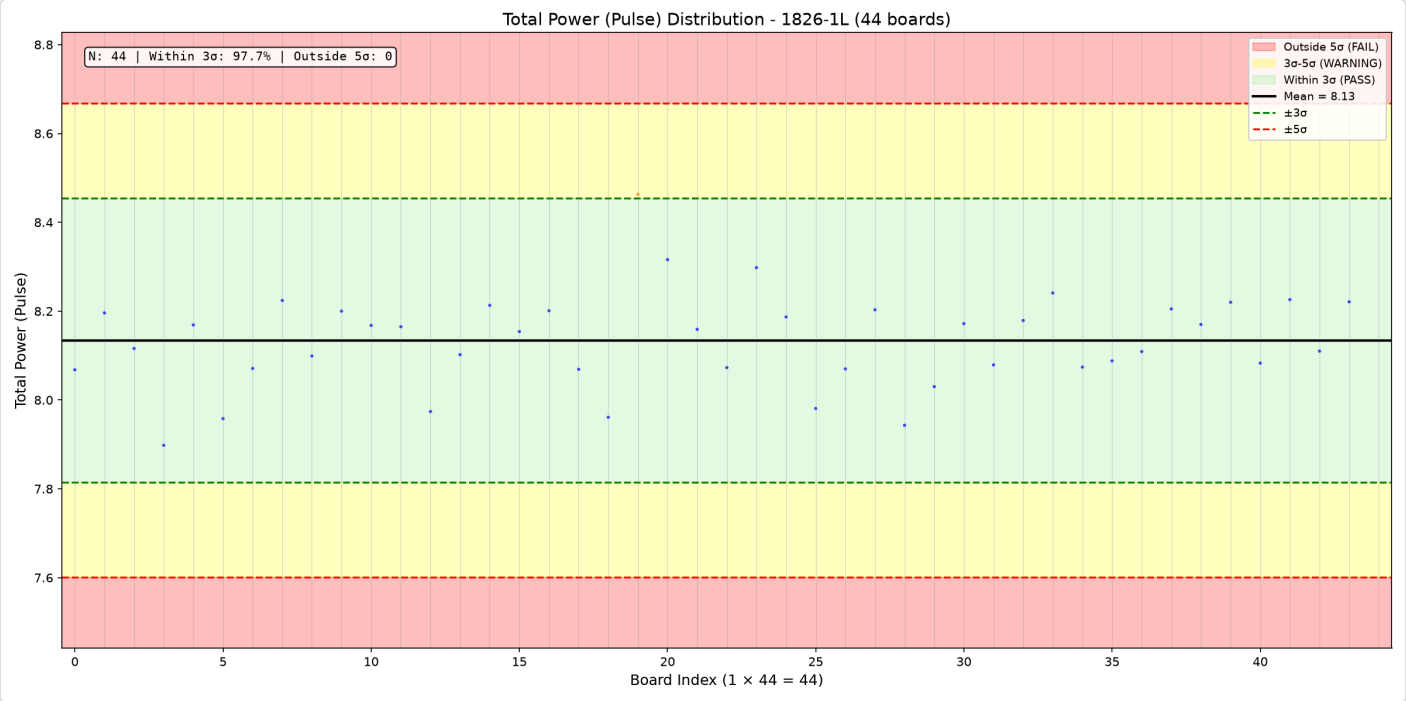


5.2 Total Power (Pulse)

Description: Total board power consumption during pulse response configuration.

Statistic	Value (W)
Mean	8.1342
Std	0.1066
5σ Lower	7.6011
5σ Upper	8.6674

Interpretation: Pulse configuration draws ~0.7W more than pedestal due to active signal processing.



6. ASIC-Level Parameters (8-Chip)

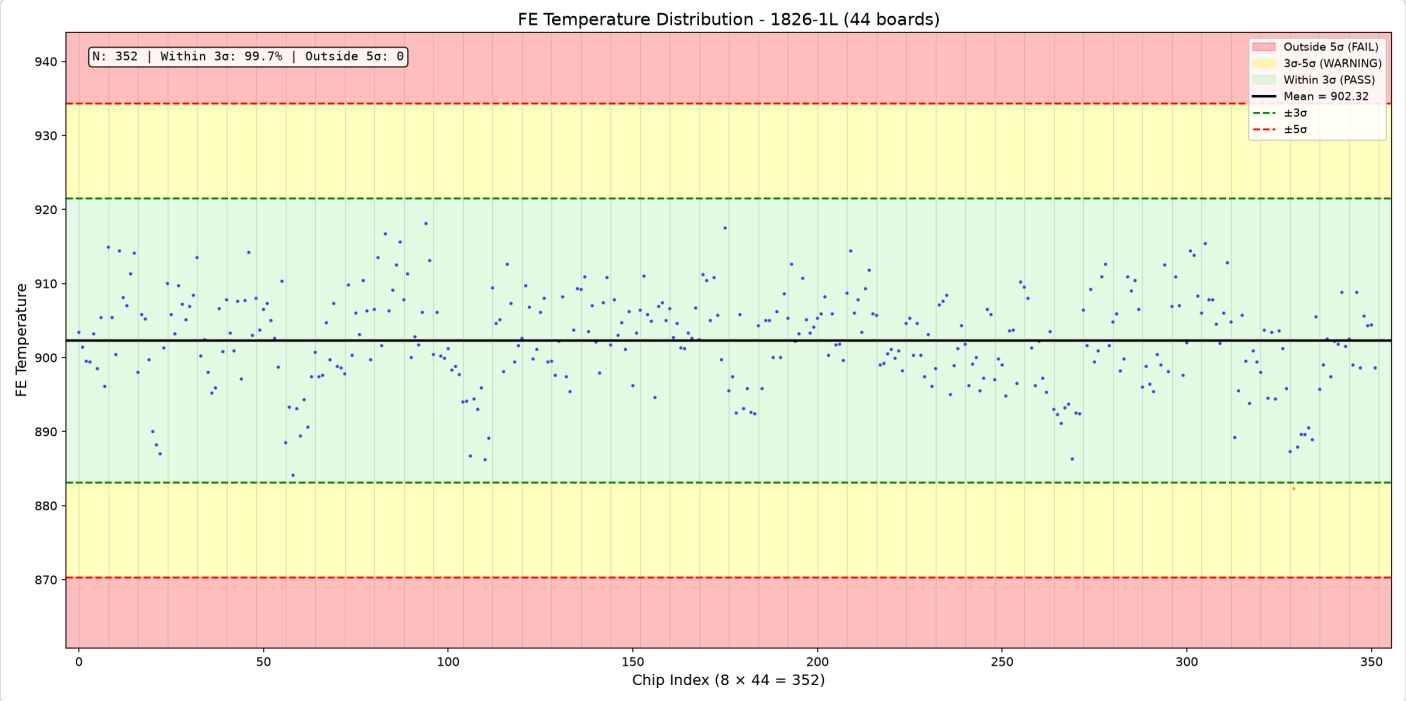
These parameters are measured per ASIC chip (8 chips per board, 352 total data points = 44 boards × 8 chips).

6.1 FE Temperature (FE T)

Description: LArASIC temperature sensor reading in ADC counts. Higher values indicate higher temperature.

Statistic	Value
Mean	902.32
Std	6.40
5σ Lower	870.33
5σ Upper	934.31

Interpretation: Temperature should be consistent across chips and boards at room temperature testing.

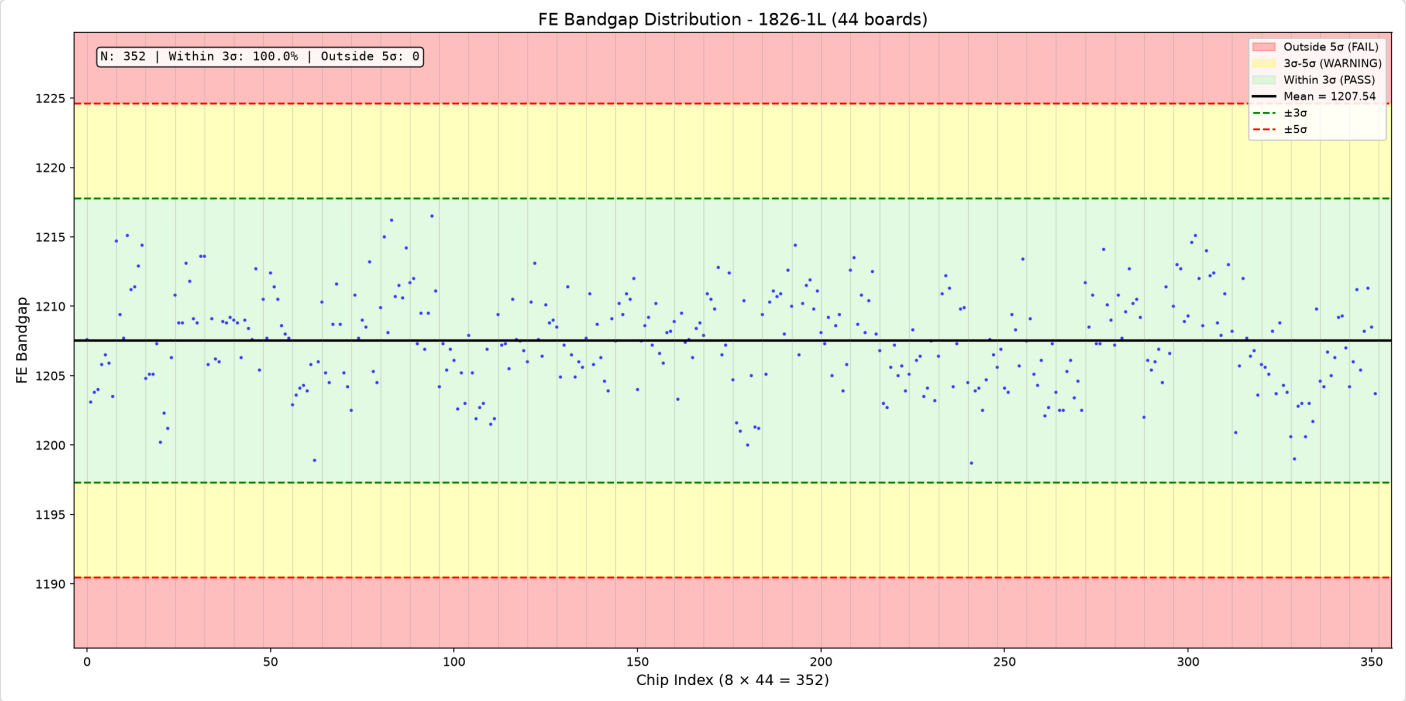


6.2 FE Bandgap (FE BGP)

Description: LArASIC bandgap reference voltage in ADC counts. Critical for stable operation.

Statistic	Value
Mean	1207.54
Std	3.41
5σ Lower	1190.48
5σ Upper	1224.61

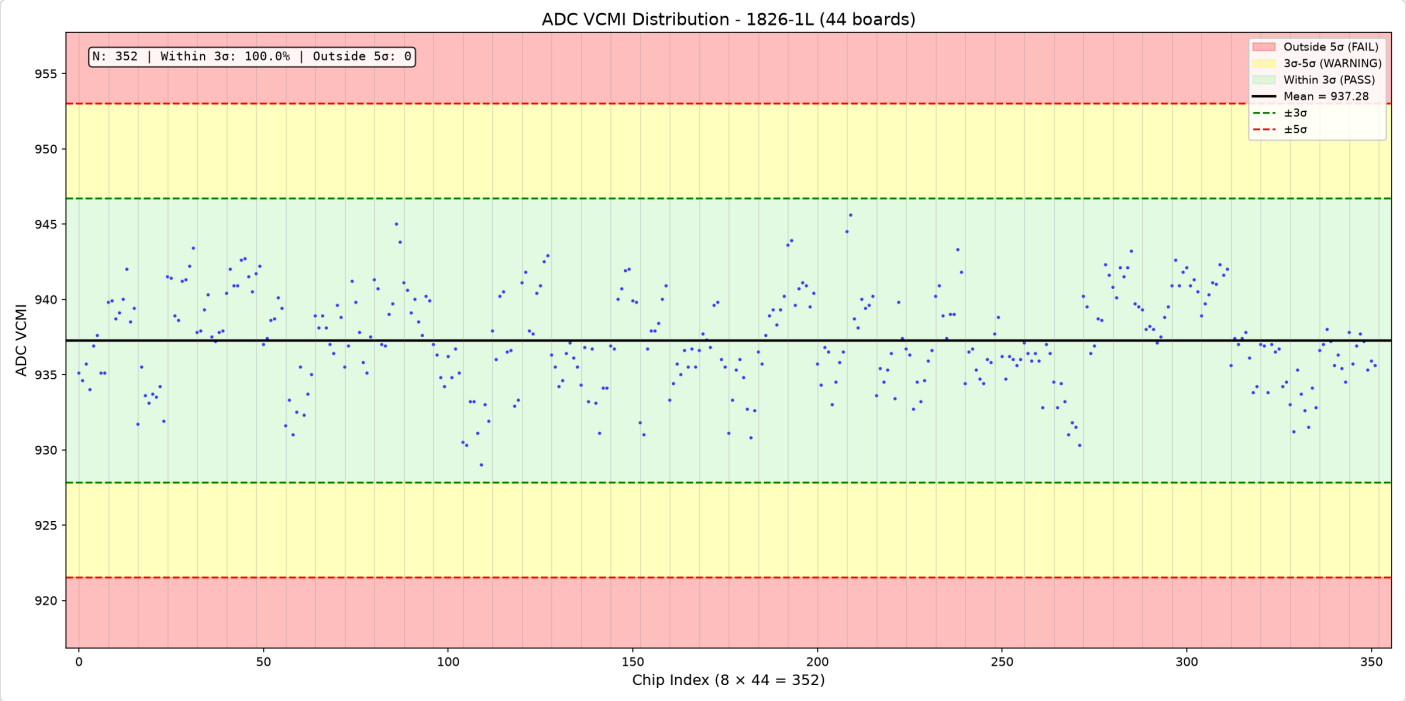
Interpretation: Bandgap provides voltage reference for the ASIC. Should be very stable.



6.3 ADC VCMI

Description: ColdADC common-mode input voltage reference.

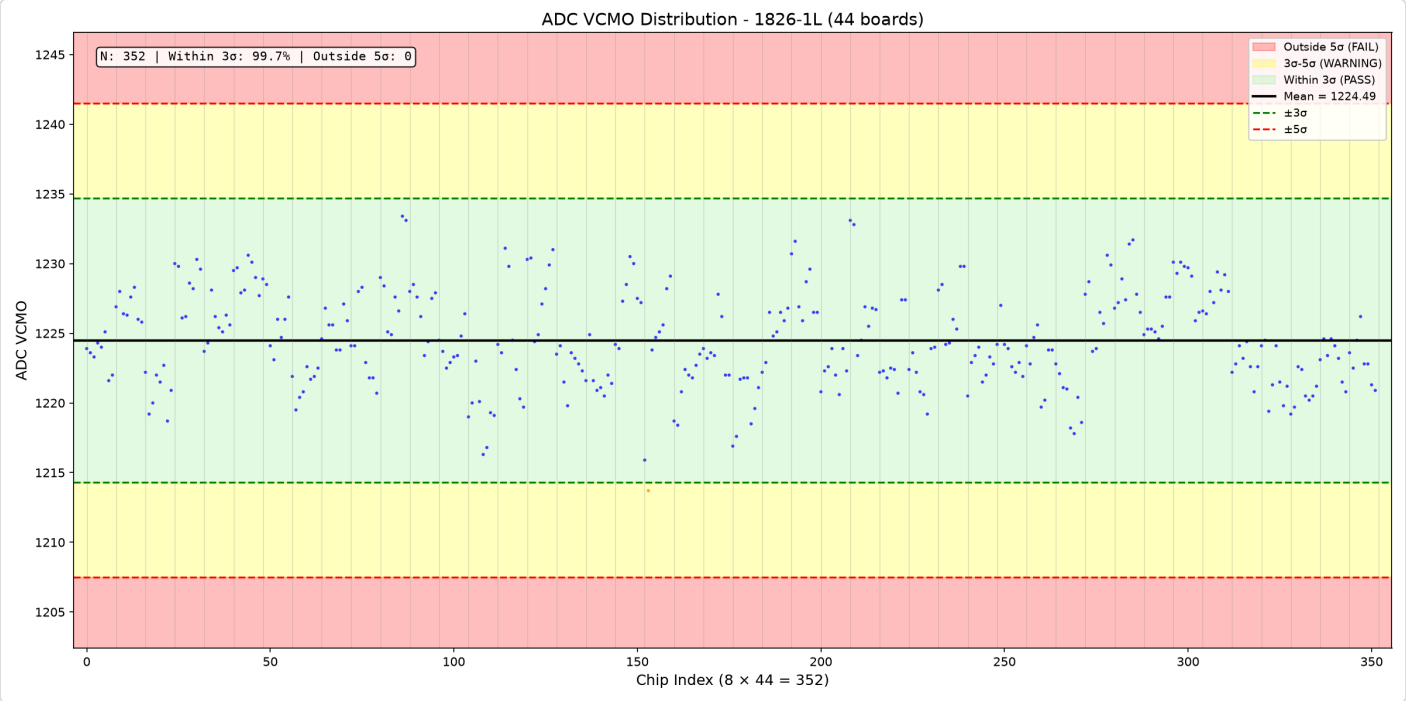
Statistic	Value
Mean	937.28
Std	3.15
5σ Lower	921.54
5σ Upper	953.01



6.4 ADC VCMO

Description: ColdADC common-mode output voltage reference.

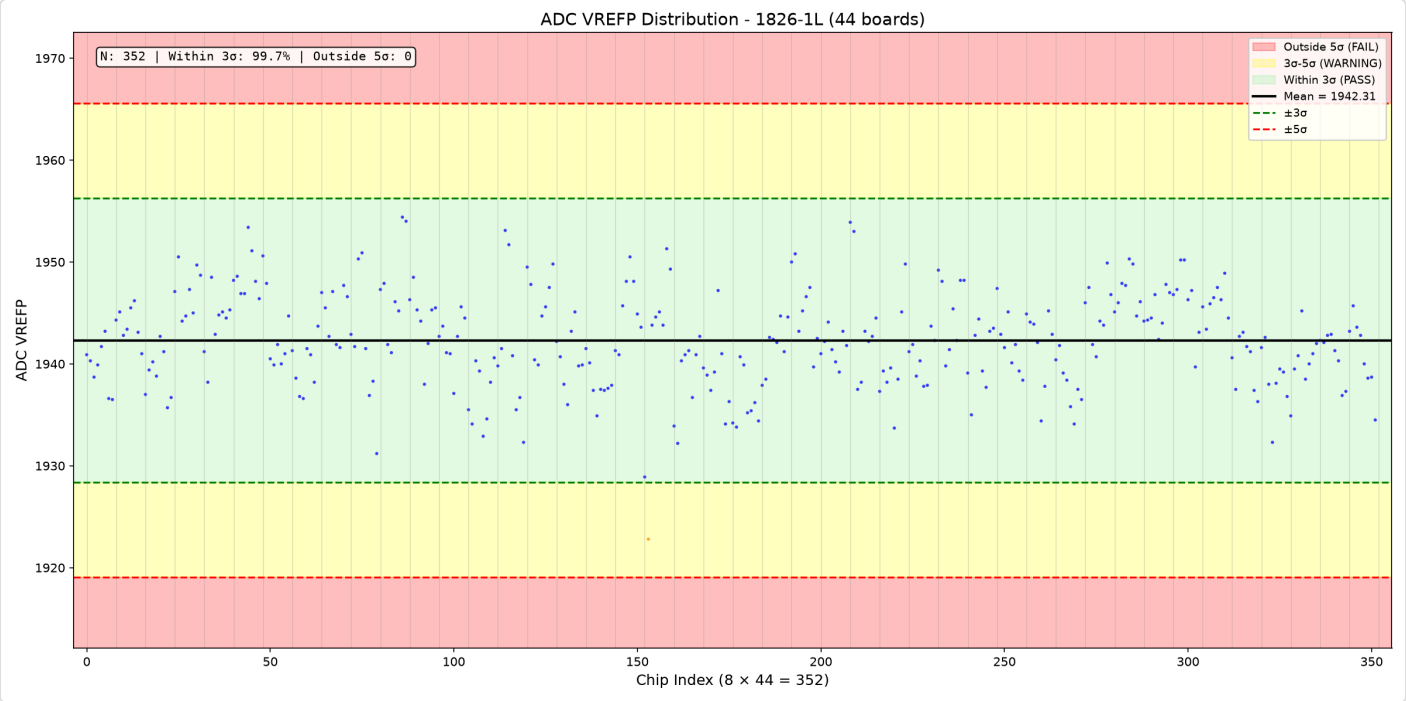
Statistic	Value
Mean	1224.49
Std	3.40
5σ Lower	1207.50
5σ Upper	1241.49



6.5 ADC VREFP

Description: ColdADC positive reference voltage. Sets the upper limit of ADC range.

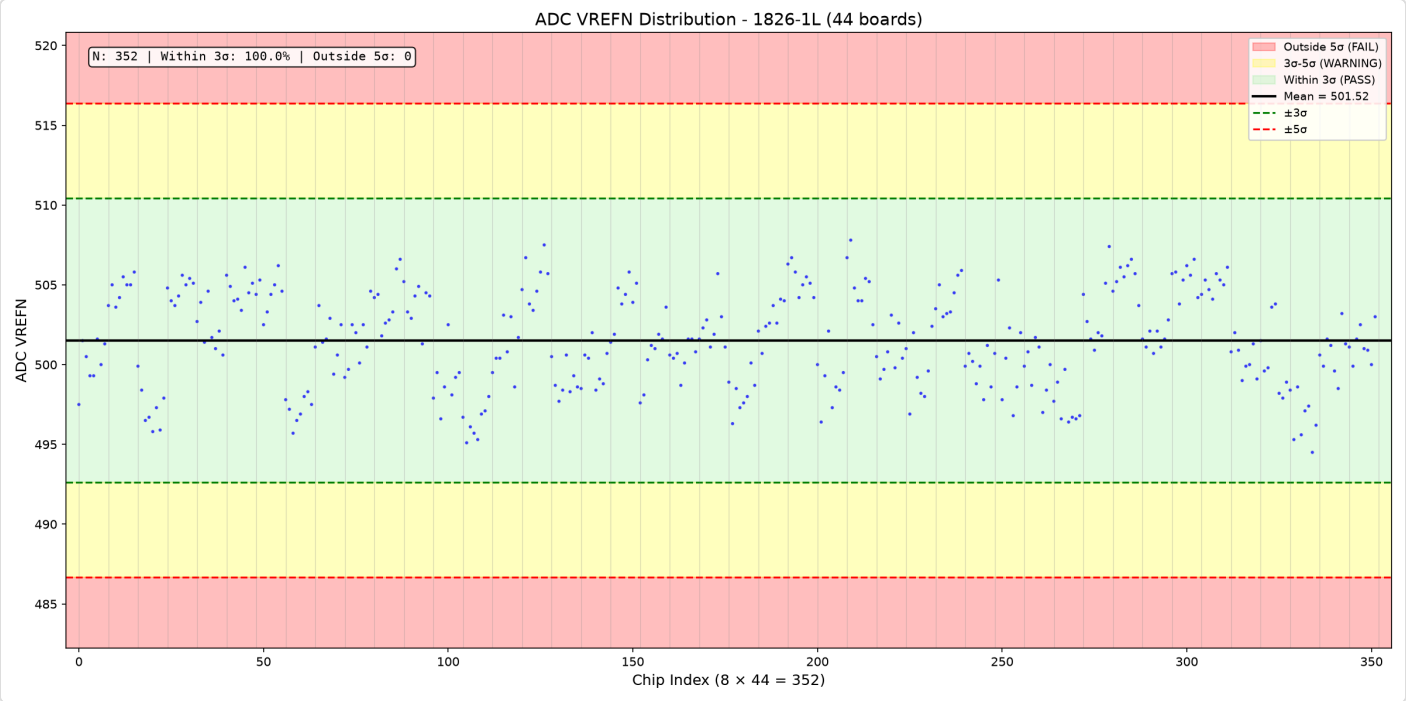
Statistic	Value
Mean	1942.31
Std	4.65
5σ Lower	1919.07
5σ Upper	1965.55



6.6 ADC VREFN

Description: ColdADC negative reference voltage. Sets the lower limit of ADC range.

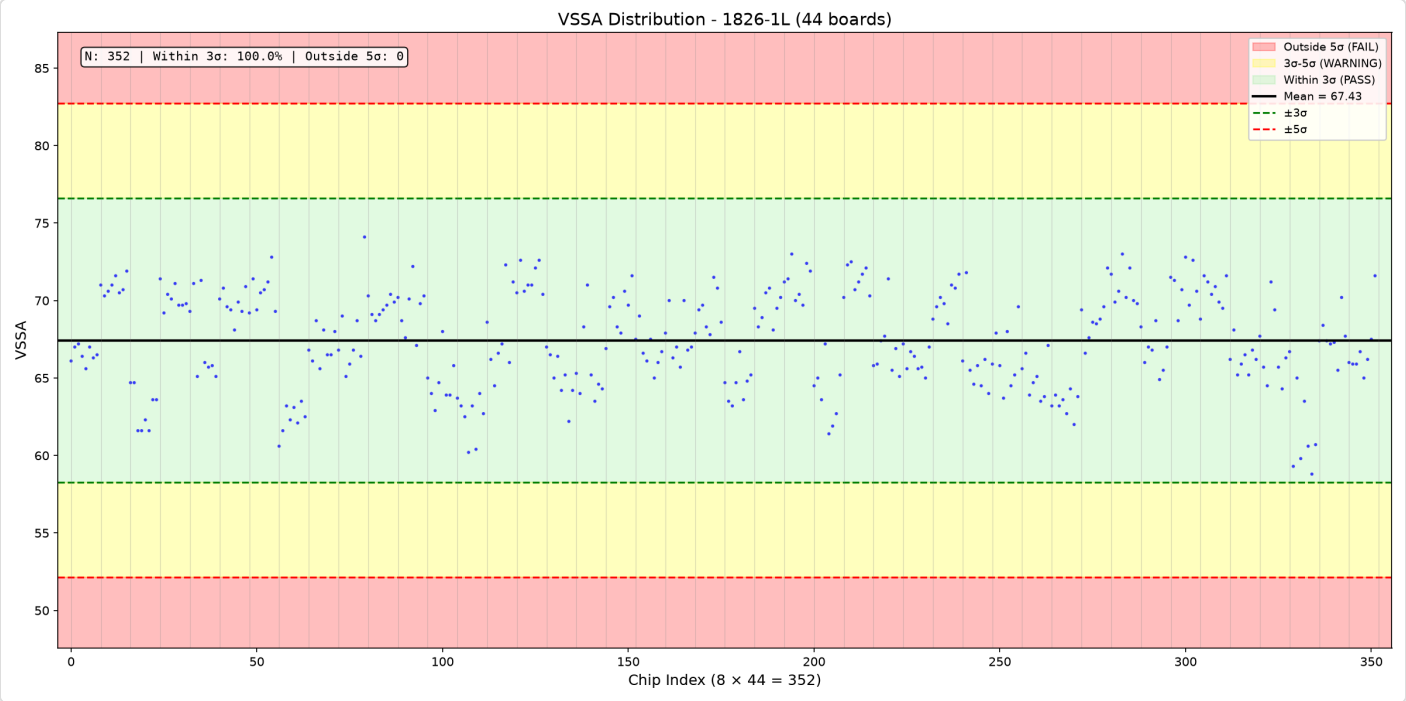
Statistic	Value
Mean	501.52
Std	2.97
5σ Lower	486.68
5σ Upper	516.36



6.7 VSSA

Description: Analog ground reference voltage. Should be near zero.

Statistic	Value
Mean	67.43
Std	3.06
5σ Lower	52.15
5σ Upper	82.72



Appendix A: Pass/Fail Criteria Summary

For QC purposes, use the **5σ bounds** as pass/fail thresholds:

Parameter	5σ Lower	5σ Upper	Unit
ped	387.81	1435.80	ADC
rms	8.93	13.47	ADC
PPK_mean	12276.23	13219.08	ADC
NPK_mean	3983.41	4839.64	ADC
BL_mean	8229.00	8930.45	ADC
Total Power (Pulse)	7.60	8.67	W
I_LArASIC (Pulse)	0.648	0.699	A
I_ColdADC (Pulse)	1.450	1.751	A

Appendix B: Board List

44 boards were used in this analysis (latest passing test only):

See `good_boards_list.txt` for the complete list of board serial numbers.

Appendix C: Excluded Boards

- **Board 00046:** Manually excluded (known bad board)
- Boards with only failed tests were automatically excluded

Report generated from QC checkout data analysis BNL Cold Electronics Testing Facility